



**Verified Carbon
Standard**

SOC SON WASTE TO POWER PLANT PROJECT



Document prepared by Energy and Environment Consultancy
Joint Stock Company (VNEEC)

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Soc Son waste to power plant project (hereinafter “the project”) includes the construction and operation of a waste treatment plant, which is located at Nam Son waste treatment complex area, Soc Son District, Hanoi. The project is owned by Ha Noi Thien Y Environmental Energy Joint Stock Company, which was established from 3 investors including Soc Son Renewable Energy Joint Stock Company, Perfect Wave Holdings Limited Company and Europe Tianying BVBA Company.

The purpose of the project activity is to treat 4,000 tons of MSW per day to generate electricity with the maximum capacity of 75 MW. However, the plant is designed and installed with the capacity of 90 MW for the following reasons:

- The nature of the waste to energy (WtE) plant is unstable operation due to the harsh working conditions. The equipment requires more frequent maintenance/repair while the maintenance/repair time is relatively long compared to other conventional thermal power plants.
- The waste treatment plant must be operated continuously. The reduction of operating capacity (partial shutdown of the plant for maintenance) will cause garbage congestion and environmental problems.

Therefore, to ensure the waste treatment capacity as well as power generation capacity, the WtE plant shall be able to have a backup capacity. The installation of 3 turbines/generators with 30 MW per unit has considered the back-up capacity for the closing of one turbine/generator to maintain or repair. In this case, the aim of the back-up design is to ensure that the two remaining turbines/generators (60 MW) are still capable of generating up to 80% of the required capacity of 75 MW. According to the FSR which was designed by the third party and also approved the national authority, with 4,000 tons of MSW per day, the maximum electricity capacity of the plant is 75 MW. Moreover, based on the signed power purchase agreement (PPA) between the PO and EVN, the maximum electricity capacity exported to the national grid is 75 MW. The electricity capacity will be capped at 75MW by both parties EVN and the project owner through the Supervisory Control and Data Acquisition (SCADA) system during operation. Therefore, the actual capacity of 75 MW is used in the financial analysis.

The annual estimated electricity output of the project activity of 388.05 GWh is supplied to the national grid via the transmission line. The Soc Son waste to power plant will use 5 Waterleau continuous grate-type incinerators for treating MSW and a combination of 5 boilers, 3 steam turbines and 3 generators to recover heat from combustion of MSW for generating electricity.

Scenario existing prior to the implementation of project activity:

Prior to the implementation of the project activity, MSW treatment in Viet Nam was mainly landfilling without any methane recovery system for flaring or energy purpose, therefore methane emissions from anaerobic decay of waste would have been released into the atmosphere.

MSW of Ha Noi is mainly treated at Nam Son landfill site in Nam Son waste treatment area and Xuan Son landfill site in Xuan Son waste treatment area, which are operated by URENCO Ha Noi – a public entity. If there is no project activity, the MSW will continue to be disposed at Nam Son landfill site. Nam Son landfill site was a sanitary landfill, operated in 2000. Nam Son solid waste treatment complex area spans over 157 ha, its construction is divided into 2 phases. Phase 1 of 83 ha contained 10 cells and was closed. Phase 2 of more than 73 ha includes 8 cells and technical infrastructure items. After many expansions and phases, up to now, this landfill receives about 5,000 tons of MSW/day, accounting for 77% of MSW generated by Ha Noi. According to Decision No.609/QĐ-TTg issued by the Prime Minister dated April 25, 2014, on Approving the master plan on solid waste disposal of Ha Noi capital to 2030, with a vision to 2050, the Nam Son landfill is planned to expand its capacity to 257 ha by 2030 and 280 ha by 2050¹. Nam Son landfill was constructed with a depth of greater 9 m, and equipped with mechanical compacting, levelling of the waste, and cover material of soil². There is one LFG recovery project (at closed cells in Nam Son landfill) that was implemented under Clean Development Mechanism (CDM)³. The LFG capture system was however abandoned due to the low price of Certified Emission Reductions (CERs) in the market.

For electricity component, prior to the implementation of the project activity, the electricity in Viet Nam was produced mainly from fossil fuel sources and solely distributed to customers through the unique national electricity grid managed by the Vietnam Electricity (EVN).

Baseline Scenario:

The baseline scenario of the project activity is the same as the scenario existing prior to the start of implementation of the project activity. The project activity is the installation of a Greenfield MSW treatment plant, the baseline scenario is disposal of the fresh waste in a solid waste disposal site (SWDS) without an LFG capture.

For electricity generation, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

¹ Decision No. 609/QĐ-TTg approving the Master Plan on solid waste disposal of Hanoi capital to 2030, with a vision to 2050

² <http://vusta.vn/chitiet/tin-tuyen-sinh-dao-tao/Rac-va-van-de-xu-ly-nuoc-ri-rac-o-Ha-Noi-1011>

³ <https://cdm.unfccc.int/Projects/DB/RINA1274714283.4/view>

Estimated GHG emission reductions:

The project activity helps to avoid methane release from SWDS(s) in the absence of the project activity and generate renewable electricity, which displaced part of the electricity otherwise supplied by grid connected fossil fuel fired power plants. Thus, GHG emission reductions can be achieved. The expected CO₂ emission reduction from the proposed project is 544,012 tCO₂e annually and 3,808,082 tCO₂e in total over the first 7 years crediting period.

1.2 Sectoral Scope and Project Type

Sectoral Scope : 1 - Energy industries (renewable / non-renewable sources)
13 - Waste handling and disposal

Project Category : ACM0022 - Alternative waste treatment process - Version 03.0

The proposed project is not a group project activity.

1.3 Project Eligibility

According to VCS Standard, version 4.3, the scope of the VCS Program includes:

- 1) The seven Kyoto Protocol greenhouse gases.
- 2) Ozone-depleting substances (ODS).
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.
- 4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.

The project activity is to produce electricity from MSW that would have been left to decay anaerobically in waste landfills. The project will result in avoidance of quantity of methane (CH₄) that otherwise would have been released into the atmosphere due to the anaerobic decay of organic waste in landfills. Beside, the project will generate renewable electricity, which displaced part of the electricity otherwise been supplied by grid connected fossil fuel fired power plants generating mainly CO₂ emission. Thus, GHG emission reductions can be achieved. This project type is not excluded from the Scope of VCS program as indicated in Table 1: Excluded project activities of VCS Standard, Version 4.3. Therefore, the project activity is eligible to develop and register under the VCS Program.

1.4 Project Design

The project activity involves the construction and operation of Soc Son Waste to Power Plant with the MSW treatment capacity of average 4,000 ton per day (equivalent to 1,460,000 tons per

year) and total electricity installed capacity of 90 MW (maximum electricity generation capacity to the national grid of 75 MW). The main equipment of the project activity includes 5 Waterleau continuous grate-type incinerators for treating MSW (5 x 800 tons of MSW per day) and a combination of 5 boilers (5 x 65.86 ton of steam per hour), 3 steam turbines and 3 generators (3 x 30 MW).

Eligibility Criteria

The proposed project activity is a single installation activity, not a group project. Thus, this section is not applicable.

1.5 Project Proponent

Organization name	Ha Noi Thien Y Environmental Energy Joint Stock Company
Contact person	Li Ai Jun
Title	Deputy General Director
Address	Nam Son Waste Treatment Complex Area, Nam Son commune, Soc Son district, Ha Noi, Viet Nam
Telephone	+ 84 24 3200 5850
Email	thiendoanhvietnam@gmail.com
Organization name	Swiss Carbon Value Ltd.
Role in the project	VCU Purchaser
Contact person	Mr. Renat Heuberger
Title	CEO
Address	Technoparkstrasse 1, Zurich, Switzerland
Telephone	+41 43 501 3550
Email	registration@southpole.com

1.6 Other Entities Involved in the Project

Organization name	Energy and Environment Consultancy Joint Stock Company
Role in the project	Project Consultant

Contact person	Ms. Dang Thi Hong Hanh
Title	Co-Founder, Deputy Executive Director
Address	8 th Floor, Diamond Flower Tower, No. 48 Le Van Luong street, Thanh Xuan district, Hanoi, Viet Nam
Telephone	+84 24 6666 9753
Email	registration@eec.vn

1.7 Ownership

As per VCS Program Definitions, version 4.2, the project ownership is the legal right to control and operate the project activities.

Ha Noi Thien Y Environmental Energy Joint Stock Company is the project owner of Soc Son Waste to Power Plant Project. They have the legal right to control and operate and get any revenue from the project activity.

The business license, project investment license, MSW treatment contract, Power Purchase Agreement (PPA) and other documents as mentioned in Section 1.14 are evidences for the ownership of the plant, equipment and process that generates GHG emission reductions.

Swiss Carbon Value Ltd. is a VCU purchaser generated from Soc Son Waste to Power Plant Project, which is demonstrated via Emission Reduction Purchase Agreement. Swiss Carbon Value Ltd is also responsible for investment and management of the project development under VCS.

1.8 Project Start Date

24/07/2022 (operation date of the proposed project activity).

1.9 Project Crediting Period

This project adopts renewable crediting periods of 7 years which can be renewed for maximum 2 times.

The first crediting period is 7 years from 24/07/2022 to 23/07/2029 (both days included).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale
Project

Large project	✓
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Year	Estimated GHG emission reductions or removals (tCO ₂ e)
24/07/2022 – 23/07/2023	195,589
24/07/2023 – 23/07/2024	384,553
24/07/2024 – 23/07/2025	513,739
24/07/2025 – 23/07/2026	602,687
24/07/2026 – 23/07/2027	664,502
24/07/2027 – 23/07/2028	707,981
24/07/2028 – 23/07/2029	739,032
Total estimated ERs	3,808,082
Total number of crediting years	7
Average annual ERs	544,012

1.11 Description of the Project Activity

Prior to the implementation of the project activity, the same amount of MSW would be disposed at Nam Son landfill site. Nam Son landfill site was a sanitary landfill, operated in 2000. Currently, the landfill receives around 5,000 tons of MSW/day. As per Decision No.609/QĐ-TTg on Approving the Master Plan on solid waste treatment of Hanoi to 2030, with a vision to 2050 dated 25 April 2014, the Nam Son landfill is planned to expand its capacity to 6,000 tons per day by 2030 and 7,000 tons per day by 2050 (corresponding to an area of 257ha and 280ha by 2030 and 2050, respectively). The project activity is expected to use 4,000 tons of MSW/day. Therefore, the Nam Son landfill area is sufficient capacity to dispose waste with a comparable annual waste acceptance rate and with the same operating lifetime as the project activity.

For electricity components, prior to the implementation of the project activity, the same amount of electricity would have been generated by the grid-connected power plants, which was mainly from fossil fuel-based power plant. The detailed information is mentioned in section 1.1 and 3.2 on baseline scenario.

The greenfield project activity involves the introduction of the grate incineration technology to incinerate MSW at Hanoi city that would otherwise be disposed of at a local landfill site. The main equipment in the project activity includes 5 incinerators (800 ton of MSW per day), 5 boilers (65.8 ton per hour), 3 steam turbines and 3 generators (30 MW x 3 units), the transformer station and electricity transmission line, exhaust gas cleaning systems and ash remove systems.

After entering the treatment area through weighbridges at the input gate, MSW will be moved to 2 temporary waste bunkers, one is for 3 incineration lines, and another is for 2 remaining incineration lines. These tanks will be designed with maximum volume of 65,954 m³ to store MSW within 8-10 days. During storage in the waste bunkers, MSW will be turned and mixed to tear waste and remove water to prevent anaerobic decomposition. Moreover, the air from the

waste bunkers will be blown to incinerator and used as a primary combustion air. MSW then provided to incinerators by clamshell buckets.

Each moving grate incinerator is equipped a system, which feeding waste to grates, including a hopper, waste chute. Garbage is pushed into the combustion chamber by the system, then moved continuously and burned directly on the grate in layer by layer, drying zone, burning zone and burn out zone. The unburnt components continue to move on the grates to get out of the combustion chamber and fall into the slag collector of furnace bottom.

The heat recovered from the incineration process will be provided to boilers. The superheated steam generated from the boilers will be fed into the steam turbine to generate kinetic energy. Turbines are connected to generators to produce electricity. The electricity generated from the project activity will be exported to the national grid through the electricity transmission line.

The flue gases from the incineration process including NO_x, SO_x, HCl, heavy metal, dioxin, furan will be treated by the exhausted gases control technology. This includes combustion control, including activated carbon adsorption, semi-dry deacidification, baghouse filter. After the treated clean gas, which meets Viet Nam Environmental Quality Standard will be release to the atmosphere from chimneys. The slag will be collected and transported to brick production plant. While fly ash will be solidified, then sent to the landfill site.

The leachate generated from MSW bunkers will be collected and gone into the wastewater treatment system and is treated together with other wastewater sources by the UASB technology to meet Viet Nam Environmental Quality Standard before discharging the environment.

The layout of the plant is shown in below figure:

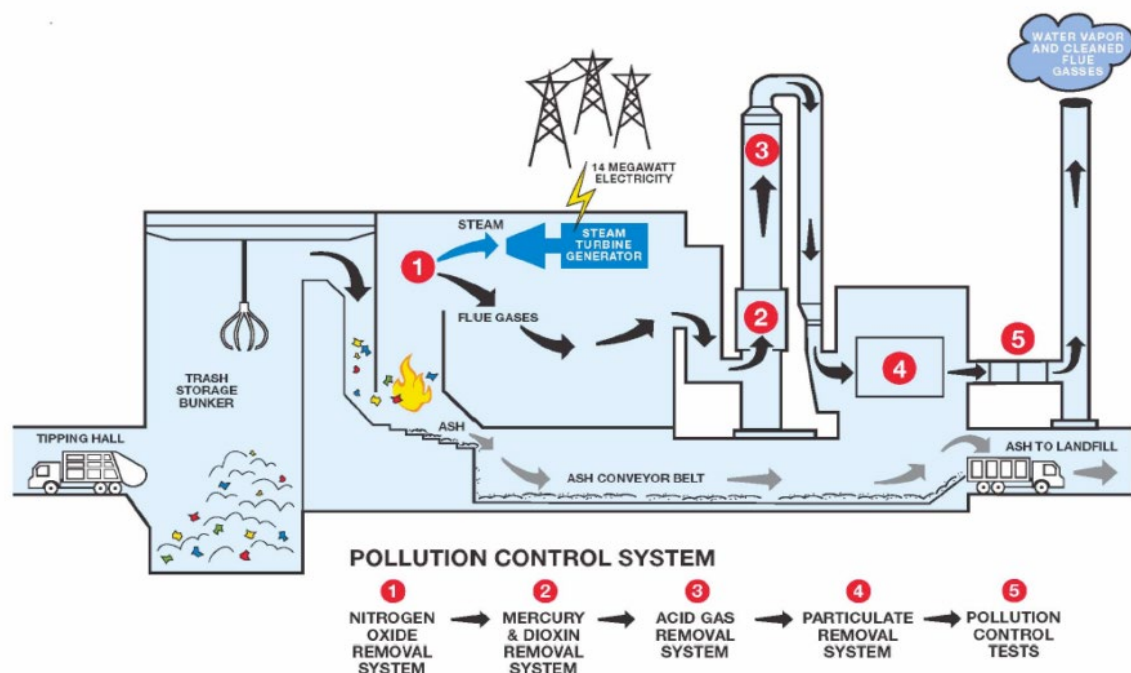


Figure 1. Line diagram of a solid waste incineration to power plant

The main technical parameters of Soc Son Waste to Power plant project are shown as table below:

Table 1. Technical parameters of proposed project⁴

Main parameters	Indicator	Values
Incinerator⁵	Type	Continuous grate
	Manufacturer	Waterleau - China
	Number	5
	Unit capacity	800 ton/day
Boiler⁶	Manufacturer	Nantong Wanda Boiler Co., Ltd., China
	Number	5
	Unit steam output	65.86 ton/h
	Rated Outlet steam Pressure	4 Mpa
	Rated Steam Temperature	400°C
	Efficiency	>80%
	Lifetime ⁷	25 years
Steam turbine⁸	Manufacturer	Nanjing Turbine & Electric Chinery (Group) Co., Ltd., China
	Type	Condensing turbine, one-body
	Model	QFW-30-2C
	Number	3
	Rated capacity	30 MW
	Speed	3000 rpm
	Lifetime ⁹	25 years
Generator¹⁰	Type	Three - phase synchronous
	Number	3
	Capacity	30 MW
	Frequency	50 Hz

⁴ Equipment supply contract

⁵ FSR, page 4-6

⁶ FSR, page 4-11

⁷ Tool 10 "Tool to determine the remaining lifetime of equipment" version 1.0

⁸ FSR, page 4-20

⁹ Tool 10 "Tool to determine the remaining lifetime of equipment" version 1.0

¹⁰ FSR, page 4-90

	Rated voltage	10.5 kV
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The technology has been imported from China. The technologies and know-how of the project have been transferred to the host country such that the labour working at the plant is local and has worked with the contractor throughout the installation and operations and maintenance phase thus resulting in the technological development of our human resource.

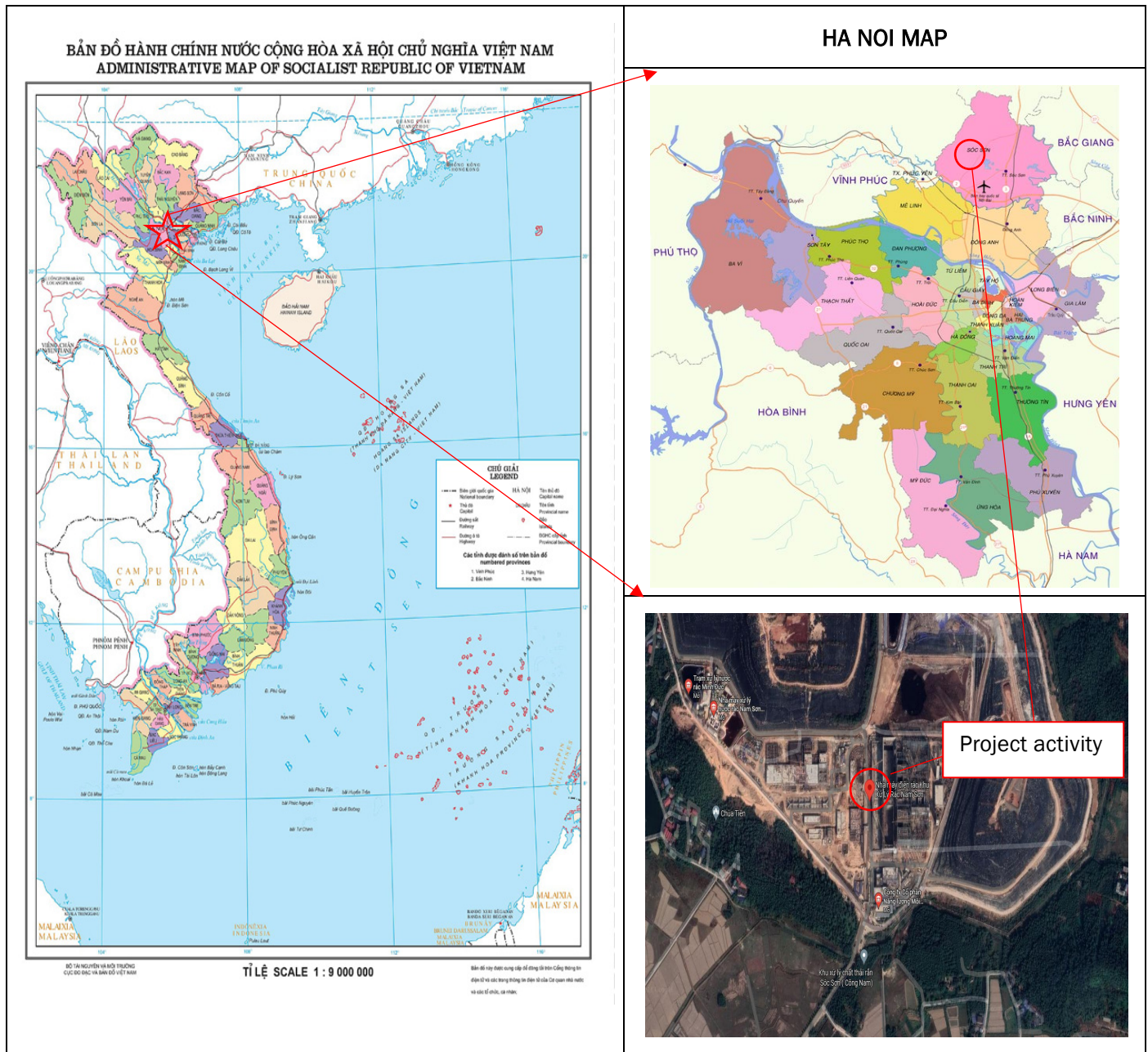
1.12 Project Location

The Project is located in Nam Son Waste Treatment Complex Area, Soc Son District, Hanoi, Viet Nam. The geo-coordinate of the project are as follows:

Table 2. Coordinates of the project activity

Co-ordinates	Soc Son Waste to Power plant
Northern latitude	21° 19' 48.0"
Eastern longitude	105° 50' 08.2"

The project location is shown in the map below:



1.13 Conditions Prior to Project Initiation

The project is a Greenfield project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site.

The baseline scenario is the same as the conditions existing prior to the project initiation; please refer to section 2.4 for further information.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project Owner has received necessary approvals for development and commissioning of the project from the State agencies and is in compliance to the local laws and regulations. This has been demonstrated through below supporting documents:

- Business license No. 0108186977 for Ha Noi Thien Y Environmental Energy Joint Stock Company issued by Investment and Planning Department of Hanoi dated on 03/03/2020, replaced from first registration dated 15/03/2018. It is compliance with the Law on enterprise No. 68/2014/QH13.
- Investment license No. 7698780162 issued by Investment and Planning Department of Hanoi dated on 15/07/2020 on certifying Ha Noi Thien Y Environmental Energy JSC is a project owner of Soc Son Waste to Power Plant, replaced from the first license dated 29/07/2017. It is compliance with Law on Investment No. 67/2014/QH13.
- Official Letter No. 1281/ TTg-CN issued by the Prime Minister dated 20/09/2018 on additional approval of Soc Son Waste to Power plant project into the Adjusted Power Development Planning VII. It is compliance with the Electricity Law and the Circular No. 43/2013/QH13 dated 31/12/2013 issued by Ministry of Industry and Trade, regulations on content, sequence, procedures for formulation, assessment, approval and adjustment to electricity development planning.
- Official Letter No. 207/HDXD-QLDA dated 10/04/2019 and No. 367/HDXD-QLDA dated 02/07/2020 issued by Department of Construction Management on approval of the Feasibility Study Report and Adjusted FSR of the project respectively. It is compliance with Law on Investment No. 67/2014/QH13.
- Decision No. 1324/QD-BTNMT dated 28/05/2019 issued by Ministry of Natural Resources and Environment (MONRE) on approval of Environmental Impact Assessment Report (EIA report) for the project activity. It is compliance with Law on Environmental Protection No. 55/2014/QH13.
- Land use right certificate issued by Hanoi People's Committee dated 11/07/2019 for Ha Noi Thien Y Environmental Energy JSC to construct and operate the plant. It is compliance with Land Law No. 45/2013/QH13.
- Power Purchase Agreement (PPA) No. 05/2019/HD-NMDR-Soc Son dated 23/05/2019 between the Vietnam Electricity and Ha Noi Thien Y Environmental Energy JSC. It is compliance with the Electricity Law.
- Municipal Solid Waste Treatment Contract No. 08/2018/HĐVXLCTSH dated 09/08/2018 between Ha Noi Thien Y Environmental Energy JSC and People's Committee of Hanoi city. It is compliance with the Environmental Protection Law.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The proposed project activity has neither been registered nor seeking registration under any other GHG programs.

The proposed project activity is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The proposed project activity is not participating in other environment credits, other GHG programs and has not been rejected by any other GHG Programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

GHG emissions reductions generated from the proposed project activity will not be used for compliance with any other emissions trading program or to meet other binding limits on GHG emissions other than VCS Program..

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of environmental credits, including renewable energy certificates.

1.17 Sustainable Development Contributions

The project contributes to the sustainable development on the local, regional, national scale in many aspects of environment, Sociality and Economy, as follows:

Environmental benefits:

- Instead of anaerobic decay of biodegradation waste that would have occurred in landfills, resulting in methane generation. The project activity will contribute to mitigation of greenhouse gas (GHG) emissions through incineration of the waste. The annual average expected emission reduction from the project activity is 544,012 tCO₂e, which contributes to SDG 13 “Respond in a timely and effective manner to climate change and natural disaster” as per mentioned in Resolution on sustainable development No 136/NQ-CP dated 25/09/2020 of The Vietnam Government. Emission reduction will be calculated at each monitoring period.
- The Project activity would reduce the amount of waste going into the dumping site thus increasing the lifetime of the waste landfill sites. The purpose of this project activity is to treat average 4,000 ton of municipal solid waste (MSW) per day. Total annual amount of municipal solid waste treated is estimated 1,460,000 ton, increasing quantity of waste at city treated by modern technology, preventing from disposing at landfill sites, which contributes to Goal of SDG 11 “Promote sustainable and resilient urban and rural development; ensure safe living and working environments; ensure reasonable distribution of population and work force by region” as per mentioned in Resolution on sustainable development No 136/NQ-CP dated 25/09/2020 of The Vietnam Government. The quantity of municipal waste treated will be measured continuously by weighbridge and gathered at each monitoring period.

- Contributing to prevent the soil, air and water pollution and improve Environment quality for local residents.
- Through recovering heat from incineration and generating clean power, the project also contributes to mitigate exhausted gases (GHG included), which would otherwise have been released into atmosphere because of production of electricity from fossil-fuel power plants.

Economic benefits:

- During the operation phase, the project activity is estimated to supply 388.05 GWh of renewable energy to the national grid annual, which contributes to SDG 7 “Ensure access to sustainable, reliable and affordable energy for all” as per mentioned in Resolution on sustainable development No 136/NQ-CP dated 25/09/2020 of The Vietnam Government.
- Advantage of incineration is that treat MSW thoroughly, minimize volume of waste, save land area to use other economic purposes instead of MSW landfill treatment.

Social benefits:

- Introducing the advanced waste to power technology on solid waste treatment that is not yet commonly applied in Viet Nam.
- Creating jobs during the construction and the operation phase and contribute to improving income for people. During the operation phase, the project activity is estimated to create approximately 180 jobs, which contributes to SDG 8 “Ensure sustained, inclusive and sustainable economic growth; full and productive employment and decent work for all” as per mentioned in Resolution on sustainable development No 136/NQ-CP dated 25/09/2020 of The Vietnam Government. List of employees at the plant will be monitored and provided at each monitoring period.
- Improving the technical skills of workers via trainings to operate the waste treatment plant.

1.18 Additional Information Relevant to the Project

Leakage Management

According to the applied methodology of ACM0022, version 03.0, no leakage is considered for incineration of MSW.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information is required.

2 SAFEGUARDS

2.1 No Net Harm

The “Environmental Impact Assessment Report” (EIAR) on 04/2019 of the project activity has been approved by MONRE via issuing Decision No. 1324/ QD-BTNMT on the Approval of EIAR dated 28/05/2019 for the Soc Son waste to power plant.

All potential negative impacts in construction and operation phase will be strictly applied mitigation measures to minimize the effect to the environment. Detail information is described further in section 2.3 below.

Therefore, there will be no harm identified from the proposed project activity.

2.2 Local Stakeholder Consultation

During implementation process, project owner conducted two meetings for local stakeholder consultation on socioeconomic and environmental impact assessment.

- The first LSC meeting for the proposed project activity was organized on 20/06/2018 at Communal Hall, Nam Son commune, Soc Son district, Ha Noi. Before the meeting was taken place, representative of local authority, impacted local people were informed and invited to the meeting through public speaker, personal invitations dated 14/06/2018.

The participants of the meeting include as follows:

- Local authorities: Representative of Nam Son Commune People’s Committee; representative of Villages of commune.
- Representative of the project participants
- Communities: 37 people including farmers (local people); local NGO such as Youth Union, Farmer Association, Fatherland Front, Woman’s Union.

- The second meeting was organized on 04/10/2019 at Communal Hall, Nam Son commune, Soc Son district, Ha Noi. The invitation letters also were sent to representative of local authority, impacted local people one week before the meeting was taken place. There are 15 participants attended the meeting.

Representatives of local authority and local people raised some comments to the project activities as follows:

Participants	Comments	Response of PO
The first meeting 20/06/2018		
Local people	- Incineration of MSW is better for environment than landfill, but need to prevent negative impacts of dust, smog.	Thanks to consensus and support of participants. - PO committed to conduct mitigation measures as

	- Request PO to manage strictly incineration process, conduct mitigation measures to prevent from pollution of environment of air, water, land. - Remind PO to consider on consumption place of brick from bottom ash.	listed in EIA report in order to prevent environmental pollution during construction and operation phase. - An automatically monitoring system of wastewater and exhausted gases will also be installed as per Viet Nam environmental regulation.
Representative of Local authority	- The project is a new project with advanced technology. The local authority is excited and look forward to the project. - Request PO to minimize pollution of dust, smog; install automatically environmental monitoring system according to environmental regulation. - This is a feasible project. Hope the people support the project	
The second meeting on 04/10/2019		
Representative of local people	- Fully support to construct the plant. Request the company to construct quickly, the plant will be operated soon in order to reduce the load of MSW delivering Nam Son landfill.	- Thanks to consensus and support of participants. - The Company committed to minimize negative impacts. - The company also were constructing to put the plant into operation on time.
Local people	- Construction phase may emit dust and noise. Request the company to implement measures mentioned as spraying water at construction area, covering vehicles loading materials. - Looking forward to the plant operate soon. - Happy that the modern plant was coming into operation. Request the Company to construct the plant ensuring quality, limit gases emission.	

The parties all consensus and support to construct the plant at Nam Son solid waste treatment complex zone, Nam Son commune, Soc Son district, Hanoi.

The minutes of meeting and the list of attendances will be submitted to the VVB for assessment.

The mechanism for on-going communication with local stakeholders:

The mechanisms for on-going communication with local stakeholders include phone, mail and direct inquiries locally with the project's managers at the project's site, as well as Nam Son commune grievance submissions.

- Project site is the place where the local stakeholders can communicate directly (or anonymously via the comment book) with the project owner. The office of the communal People's Committee is a standing unit of the People's Committee to deal with comments from local community on all matters of the commune and is the contact point between local authority and the residents. The comment books will be securely placed in the chosen locations and daily checked by responsible persons. Administrative Department of PO will receive inputs or comments from stakeholders and forward to relevant department to corporately address and close issues.
- The telephones are located at the office of the project owner at the project's site and at the office of the VCS consultancy company to allow more practical communication with local stakeholders. There is always a receptionist on the desk to answer the calls or have the messages recorded. All received calls shall be logged and recorded in Comment Book with the date, comments, action requested and project responses. Stakeholders are not required to give their personal details when they wish to make a comment.
- The notification on continuous grievance mechanism is also public on the website of consultant unit. This is always available to get any comment from stakeholders. <https://eec.vn/thong-bao-ve-viec-tiep-tuc-tiep-nhan-y-kien-cong-dong-cho-du-an-nha-may-dien-rac-soc-son/>.

2.3 Environmental Impact

Pursuant to the Law on Environmental Protection No. 55/2014/QH13 dated 23/06/2014, Decree No.18/2015/ND-CP dated 14/02/2015 on Environmental protection planning, strategy environmental assessment (SEA), environmental impact assessment (EIA) and environmental protection plans, the EIA has been carried out Apr 2019 and already approved by MONRE via Decision No. 1324/ QD-BTNMT dated 28/05/2019.

As per the EIAR, the possible environmental impacts of the project and responding measurements taken by the project owner are summarized as follows:

Impacts and mitigation measures in the construction phase:

Dust and exhaust gases, noise

Dust, exhaust gases and noise generated from vehicles for transportation of materials, machines, equipment and construction equipment, or ground levelling activities. However, these emissions are short term and localized at the project area. The PO also conducts mitigation measures to minimize any potential native impacts as follow:

- Request contractors to implement environmental protection measures as well as supervise their compliance.
- Using advanced vehicles, machines and equipment must meet Viet Nam Standard on technical and environmental safety. They also have to be maintained periodically.

- Don't transport excess regulated weight; covering materials on transportation equipment when incoming or going out of the project site.
- Arranging operation time of vehicles and construction suitably, avoiding gather materials at the same time. Avoid construction activities during peak-hours.
- Covering and watering construction areas, to prevent from dispersion of dust into the ambient environment.
- Provide labour protection equipment such as safety masks, safety helmets, protective clothes for workers.

Wastewater

Domestic wastewater:

Domestic wastewater comes from workers' activities. To mitigate the impacts of wastewater, the following measures will be conducted by the PO:

- Increase using of local workers, install portable composite toilets. During the construction phase, the contractors will hire functional units to collect and treat waste from storage tanks of portable toilets.
- Rainfall flow in construction site, it can contain excavated soil or waste, then it will be preliminarily deposited at Pits before coming out the environment.
- Prohibit washing equipment, machines at water sources or places which washing water can directly flow into the drainage system of the area.

Construction wastewater:

Construction wastewater from washing area, mixing materials mainly contains suspended dust and may fall into and make blockage the general drainage system. Therefore, mitigation measures will be implemented to prevent from this:

- Construction wastewater will be led to pits before flow into the general drainage system.
- Don't gather materials near the drainage system.
- Always check and dredging periodically if construction wastes fall into the drainage system.

Solid waste

The solid waste will be collected and treated by URENCO during the construction phase.

Impacts and mitigation measures in the operation phase:

Gases emission

Gases emission from the transportation of MSW, incinerators or wastewater treatment system. These emissions will be managed and handled with the following measures:

- Trucks loading MSW must be covered to avoid dropping or emitting the bad odor of MSW.

- After moving MSW to waste bunkers, trucks will be washed before coming out of the plant.
- Periodically spraying deodorant chemical to minimize bad odor, bacteria.
- Plant trees around the plant to improve the microclimate.
- For flue gas, they will be treated in gas emission treatment system including the De-NO_x system (SNCR - Selective non-Catalytic Reduction), acid removal tower, dioxin and heavy metal removal by activated carbon, bag filters to extract dust from gas stream and chimney system. The continuous monitoring system will be installed to monitor main pollutants in flue gas (NO₂, SO₂, CO, HCl and dust) and treated gases emission need to meet the environmental quality standard before emitting the atmosphere.
- Fly ash is solidified and landfilled while bottom ash is transported to the brick production facility.

Wastewater

The wastewater from domestic activities (handled in septic tanks), leachates, washing of workshop and equipment will be led to the leachate treatment system and treated to meet the environmental quality standard before discharging receiver sources.

Solid waste

The solid waste will be gathered and moved to the waste reception area to be used as input waste of the plant.

2.4 Public Comments

This project was open for public comment from 11/08/2021 to 10/09/2021. No comments were received during the public comment period. ¹¹

2.5 AFOLU-Specific Safeguards

The project activity involves to construct a MSW treatment plant, a non-AFOLU project, thus this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Methodology : ACM0022

¹¹ <https://registry.verra.org/app/projectDetail/VCS/2567>

Title : Alternative waste treatment process
 Version : 03.0
 Reference : <https://cdm.unfccc.int/methodologies/DB/YINQ0W7SUY002S6GU8E5DYVP2ZC2N3>

Reference tools:

- Tool02 – Combined tool to identify the baseline scenario and demonstrate additionality - version 07.0¹²
- Tool03 - Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion, version 03.0¹³
- Tool04 – Emissions from solid waste disposal sites, version 08.0¹⁴
- Tool05 – Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0¹⁵
- Tool07 - Tool to calculate the emission factor for an electricity system, version 7.0¹⁶
- Tool27 – Investment analysis, version 11.0¹⁷.

3.2 Applicability of Methodology

The project activity is eligible to apply the methodology of ACM0022 – Alternative waste treatment processes – version 03.0 as discussed in table below:

Table 3. Project's applicability under the methodology ACM0022 – version 03.0

Applicability criteria	Project's characteristics meet to criteria
The methodology applies to the project activities that install and operate new plants for the treatment of fresh waste through any combination of the following processes: (a) Composting process under aerobic conditions; (b) Anaerobic digestion with biogas recovery and flaring and/or its use; (c) Co-composting of wastewater in combination with solid waste;	The project activity involves constructions of a new MSW treatment plant in Nam Son sold waste treatment complex area, Soc Son district, Hanoi, Viet Nam. It is a plant with incineration technology which is listed as the process (g) in the methodology.

¹² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-02-v7.0.pdf>

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf>

¹⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v8.0.pdf>

¹⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf>

¹⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

¹⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf>

(d) Anaerobic co-treatment of wastewater in combination with solid waste; (e) Mechanical/ thermal treatment process to produce refuse-derived fuel (RDF) or stabilized biomass (SB) that is produced within the project boundary and its use; (f) Gasification process to produce syngas and its use; (g) Incineration of fresh waste for the generation of thermal/electric energy. (h) The following conditions apply to all project activities using this methodology:	
(i) The project plant only treats fresh/ wastewater for which emission reductions are claimed, except for cases involving composting, co-composting and anaerobic digestion;	(i) The project only treats the fresh waste, from which the emission reductions are claimed.
(j) Neither fresh waste nor products from the project plant are stored on-site under anaerobic conditions;	(j) The MSW collected will be stored in the tank for less than 10 days¹⁸; furthermore, the crane will mix the waste in the tank to prevent the anaerobic decomposition. Moreover, the air from the waste bunkers will be blown to incinerator and used as a combustion air. Thus, Neither fresh waste nor products from the plant are stored on-site under anaerobic conditions.
(k) Any wastewater discharge resulting from the project activity is treated in accordance with applicable regulations;	(k) The domestic sewage and the leachate from the project will be treated in wastewater treatment system at the plant to meet Viet Nam Standard QCVN40:2011/BTNMT before discharging into the environment. The effluent of the treatment system will be measured by the third party periodically.
(l) The project activity does not reduce the amount of waste that would be recycled in the absence of the project activity. This shall be justified and documented in the	(l) Until date, domestic solid waste in Ha Noi is mainly treated via disposing at landfills (accounting for around 98% of collected MSW), and incineration without power generation (accounting for 2% of collected

¹⁸ FSR, page 4-36

clean development mechanism project design document (CDM-PDD);	MSW) ¹⁹ . Prior implementing the project activity, the waste would be disposed at Nam Son landfill. Nam Son landfill site was a sanitary landfill, operated in 2000. Currently, the landfill receives around 5,000 tons of MSW/day. As per Decision No.609/QD-TTg on Approving the Master Plan on solid waste treatment of Hanoi to 2030, with a vision to 2050 dated 25 April 2014, the Nam Son landfill is planned to expand its capacity to 6.000 tons per day and 7,000 tons per day by 2030 and 2050, respectively. Therefore, the Nam Son landfill area is sufficient capacity to dispose the amount of MSW received by the project during the project's lifetime. Without the project activity, the MSW would be buried at Nam Son landfill. It means that the project doesn't reduce the amount of waste that would be recycled in the absence of the absence of the project activity.
(m) When applicable regulations mandate any waste treatment process implemented under the project activity, the rate of compliance with such regulations for the treatment process is below 50 per cent;	(m) There are presently no regulations or policies in Viet Nam require treating waste before dumping at the landfill site.
(n) Hazardous wastes/wastewater are not eligible under this methodology.	(n) The project only treats the municipal domestic waste,
The methodology is only applicable if the baseline scenario is: (a) The disposal of the fresh waste in a SWDS with or without a partial LFG capture system (M2 or M3);	(a) The disposal of the fresh waste in SWDS without a partial LFG capture system (M3)
(b) In the case of co-composting or co-treatment of wastewater in an anaerobic digester, the treatment of organic wastewater in either an existing or new	(b) It is not applicable since the project is no co-composting or co-treatment of wastewater in an anaerobic digester.

¹⁹ <https://moc.gov.vn/vn/tin-tuc/1184/67792/ha-noi-xu-ly-chat-thai-ran-sinh-hoat---uu-tien-ung-dung-cong-nghe-tien-tien.aspx>.

aerobic lagoon or sludge pit without methane recovery (W1 or W4);	
(c) In the case of electricity generation, the electricity is generated in an existing/new captive fossil fuel fired power-only plant, captive cogeneration plant and/or in the grid (P2, P4 or P6);	(c) The electricity is generated in the grid (P6)
(d) In the case of heat generation, the heat is generated in an existing/new fossil fuel fired cogeneration plant, boiler or air heater (H2 or H4).	(d) It is not applicable since the project activity doesn't generate heat or displaces heat generation in the baseline. Please refer to section 3.4 of this PD for more details.
<i>For incineration, the applicable types of waste that may be treated should be fresh waste</i>	<i>The MSW treated is the fresh waste.</i>
<i>For incineration, applicable products and their use should be electricity and/or heat</i>	<i>The heat generated from the incinerators will be fed into the boilers to produce steam. The superheated steam from boilers is used to generate electricity by steam turbine and generator systems. Heat/ steam generated is only used for electricity generation purpose at the plant. The electricity generated by the project activity will be consumed internally at the plant, the surplus will be exported to the national grid.</i>
<i>For incineration, applicable waste by products should be:</i> <i>1. Incineration by-product (e.g. inert materials);</i> <i>2. Wastewater discharge</i> <i>3. Non-biodegradable materials that may have market value (i.e. glass, metals and plastics)</i>	<i>1. Incineration by-product: inert materials after incineration which will be landfilled.</i> <i>2. Wastewater discharge will be treated at the wastewater treatment system at the plant</i> <i>3. The waste received for incineration contains non-biodegradable materials (i.e 0.5% of glass, 0.9% of metal, 3% of plastic). Therefore, the project activity doesn't generate non-biodegradable materials as by-product from project activity.</i>
<i>For incineration, specific applicability conditions:</i>	<i>Incineration technology of the project activity is the grate type.</i> <i>Energy generated by waste:</i>

<i>Incineration technology is rotary kiln, rotating fluidized bed, circulating fluidized bed, hearth or grate type;</i> <i>The fraction of energy generated by auxiliary fossil fuels is not more than 50% of the total energy generated in the incinerator</i>	$1,460,000 * 1000 * 7,120^{20} \text{ kJ/kg} = 1.03952 * 10^{13} \text{ kJ/yr}$ <i>Energy generated by DO (auxiliary fossil fuel):</i> $1,200^{21} \text{ ton/yr} * 43^{22} \text{ GJ/t} = 51.6 * 10^9 \text{ kJ/yr}$ <i>Thus, the energy generated by DO accounts for 0.50 % of the total energy generated in the incinerators, which is far less than 50%.</i>
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The project activity is also eligible to apply the methodological tool as listed above. Justification is discussed in below:

Tool04 – Emissions from solid waste disposal sites, version 08.0:

Applicability conditions	Project's characteristics meet to conditions
The tool can be used to determine emissions for the following types of applications: (a) Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas"). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g. measuring the amount of methane captured from the SWDS)	The project activity involves constructions of a new MSW treatment plant, which avoids the disposal of waste at a SWDS. Thus, application B is applied for the project activity.

²⁰ Average NCV of waste is calculated as the average of High calorific value and Low calorific value, which sourced from FSR, page 3-6.

²¹ FSR, page 1-12

²² IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, section 1.4, no refinement, table 1.2 of volume 2, section 1.4.1.3 of IPCC 2006 Guidelines.

(b) Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	
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Tool 05_Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0

Applicability conditions	Project's characteristics meet to conditions
If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power	The project activity involves MSW incineration and electricity generation for internal use and the excess is to supply to the national grid. In case of the electricity generated from the plant is insufficient, the plant will import electricity from the national grid for internal use. Therefore, scenario A is applied to the project activity.

plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	This tool is applied to monitor amount electricity generated in the project scenario. The electricity generated from project activity is supplied to the grid. Therefore, the project activity meet scenario I in the tool.
This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO2 emissions	Not applicable

The remaining tools are applied for all types of proposed project activities and referred by the methodology.

3.3 Project Boundary

As per methodology ACM0022, version 03.0, the spatial extent of the project activity is the SWDS where the waste is disposed of in the baseline, anaerobic lagoons or sludge pits treating organic wastewater in the baseline, and the site of the alternative waste treatment process (es). The boundary also includes on-site electricity and/or heat generation and use, on-site fuel use and the wastewater treatment plant used to treat the wastewater by-products of the alternative waste treatment process(es). The project boundary does not include facilities for waste collection and transport.

In the case that the project provides electricity to a grid, then the spatial extent of the project boundary will also include those plants connected to the energy system to which the plant is connected.

The project activity involves the incineration of MSW and exports electricity to the national grid, does not claim the emission reduction from anaerobic lagoons or sludge pits treating organic wastewater. Then the project boundary will include the SWDS where the waste is disposed of in the baseline, the site of treatment process, on-site electricity and/or heat generation and use, on-site fuel use, the wastewater treatment plant, and those plants connected to the national grid. The project boundary is shown in the figure below:

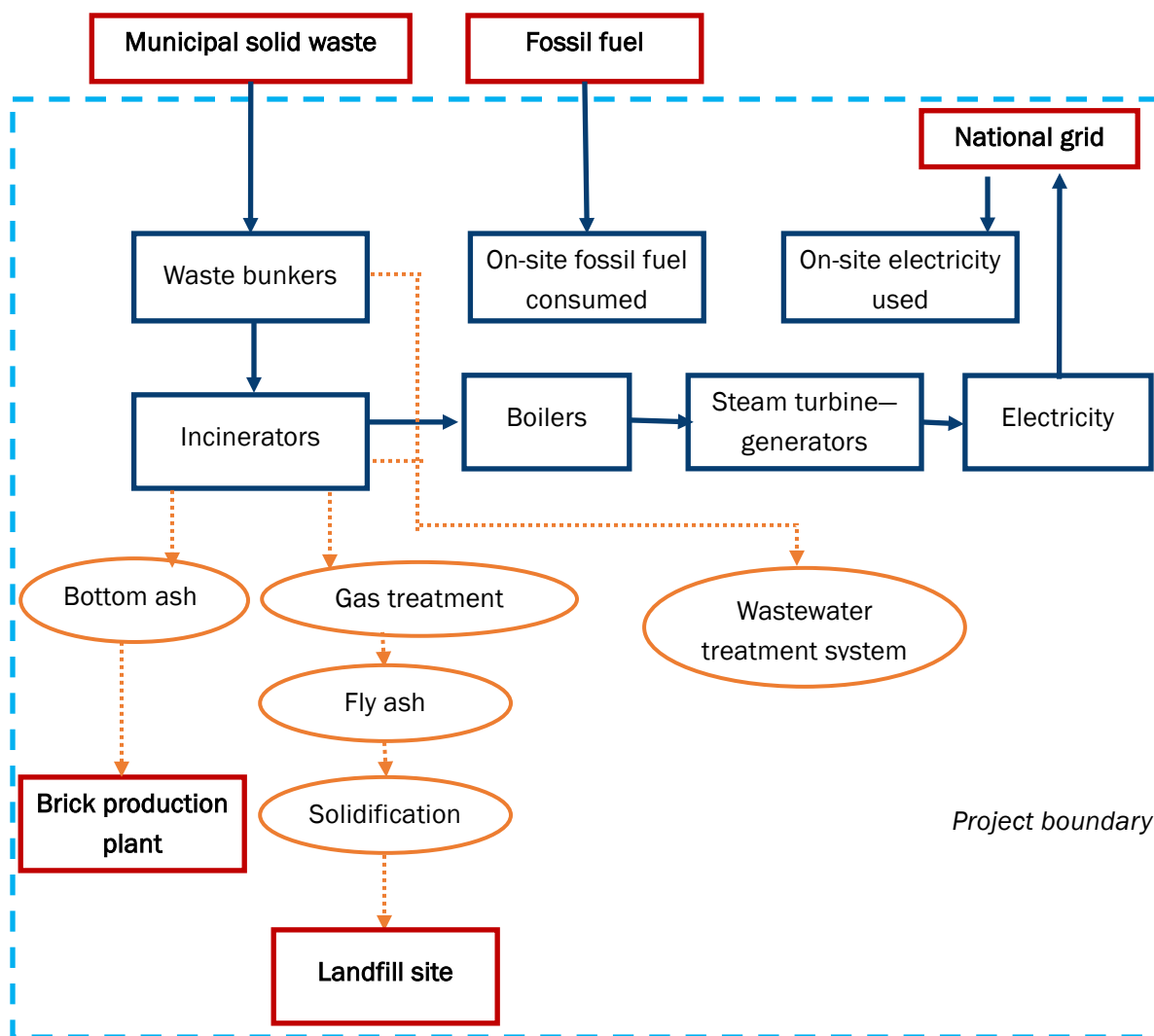


Figure 3: Project boundary

The GHGs and emission sources included in or excluded from the project boundary are shown in the Table below:

Table 4: Source and gases included or excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from	CO ₂	No	CO ₂ emissions from the decomposition of fresh waste are not accounted for

Source		Gas	Included?	Justification/Explanation
	decomposition of waste at the SWDS	CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from landfills. Exclusion of this gas is conservative
	Emissions from electricity generation	CO ₂	Yes	Major source. Electricity generation is included in electricity grid.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
Project	Emissions from on-site fossil fuel consumption due to the project activity other than for electricity generation	CO ₂	Yes	May be an important emission source. It includes auxiliary fossil fuels needed to be added into incinerator, etc.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from on-site electricity use	CO ₂	Yes	May be an important emission source. CO ₂ emissions from imported electrical power.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
	Emission from the waste treatment processes	CO ₂	Yes	CO ₂ emission generated from incineration of MSW
		CH ₄	Yes	CH ₄ emitted from incineration of MSW
		N ₂ O	Yes	N ₂ O may be emitted from incineration of MSW
	Emissions from wastewater treatment	CO ₂	No	Excluded for simplification
		CH ₄	Yes	Waste water leachate is treated anaerobically in UASB. Thus, CH ₄ emission may be emitted into the atmosphere
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be small

3.4 Baseline Scenario

According to methodology ACM0022, version 2.0, identify the baseline scenario and demonstrate additionality using the “Combined tool to identify the baseline scenario and demonstrate additionality” and following the requirements below.

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

This step is not applied to the project activity since it is not first-of-its-kind, hence the additionality of the project will be demonstrated in next steps below.

Step 1: Identification of alternative scenarios

Sub-step 1a: Define alternative scenarios to the proposed project activity:

Identify all alternative scenarios that provide the same output (service or product) as the proposed project activity. These alternative scenarios shall include:

- (a) M1: The project activity without being registered as a VCS project activity (i.e. any (combination) of the waste treatment options listed in Table 2 of methodology);
- (b) M2: Disposal of the fresh waste in a SWDS with a partial capture of the LFG and flaring of the captured LFG;
- (c) M3: Disposal of the fresh waste in a SWDS without a LFG capture system;
- (d) M4: Part of the fresh fraction of the solid waste is recycled and not disposed in the SWDS;
- (e) M5: Part of the fresh fraction of the solid waste is treated aerobically and not disposed in the SWDS;
- (f) M6: Part of the organic fraction of the solid waste is incinerated and not disposed in the SWDS;
- (g) M7: Part of the organic fraction of the solid waste is gasified and not disposed in the SWDS;
- (h) M8: Part of the organic fraction of the solid waste is treated in an anaerobic digester and not disposed in the SWDS;
- (i) M9: Part of the organic fraction of the solid waste is mechanically or thermally treated to produce RDF/SB and not disposed in the SWDS.

Assessment of potential waste management alternatives is listed as below:

Description of alternative scenarios		Plausible and credible alternative?	Rationale
M1	The project activity without being registered as a CDM project activity (i.e. any (combination) of the waste	Yes	M1 is realistic and credible since incineration is one of methods for MSW treatment in Viet Nam.

	treatment options listed in Table 2 of methodology)		
M2	Disposal of the fresh waste in a SWDS with a partial capture of the LFG and flaring of the captured LFG	No	M2 cannot be a realistic scenario since in the current situation the fresh waste's organic fraction is diverted to cells of Nam Son landfill which does not have any LFG flaring system. In Hanoi, there is only one LFG recovery project (at closed cells in Nam Son landfill) that was implemented under Clean Development Mechanism (CDM). The LFG capture system was however abandoned due to low price of Certified Emission Reductions (CERs) in the market.
M3	Disposal of the fresh waste in a SWDS without a LFG capture system	Yes	M3 is a realistic and credible scenario. The alternative doesn't require any investment. There are no mandatory regulatory or contractual requirements that mandate to undertake the alternative. Therefore, M3 is considered for further assessment. It may note that in pre-project scenario, the MSW is disposed at Nam Son landfill without a LFG capture system. Nam Son landfill site was a sanitary landfill, operated in 2000. Currently, the landfill receives around 5,000 tons of MSW/day. As per Decision No.609 on Approving the Master Plan on solid waste treatment of Hanoi to 2030, with a vision to 2050 dated 25 April 2014, the landfill is planned to expand to 280 ha, with corresponding capacity of 7,000 tons per day by 2050. Therefore, the landfill area is sufficient to dispose the same amount of waste received by the project during the project's lifetime.
M4	Part of the fresh fraction of the solid waste is recycled and not disposed in the SWDS	No	The project activity involves to sorted recyclable waste before the waste is transported to the transfer stations and moved to the plant. Part of sorted waste will be acquired by waste acquisition

			station and then finally sold to waste recycle facilities. In this sense, M4 is only a part of the project activity, therefore M4 is not considered as a realistic alternative.
M5	Part of the fresh fraction of the solid waste is treated aerobically and not disposed in the SWDS	No	M5 can not be considered as an alternative to the project activity because the project activity aims not only introducing a MSW treatment system but also generating electricity. There is no possibility to generate electricity when treating the solid waste aerobically. Thus M5 is discarded from further consideration.
M6	Part of the organic fraction of the solid waste is incinerated and not disposed in the SWDS	No	M6 is a part of the project activity and included in M1. Therefore, M6 is not considered as a realistic alternative.
M7	Part of the organic fraction of the solid waste is gasified and not disposed in the SWDS	No	Until date, there is no gasification project in Viet Nam ²³ . It is not business as usual in Viet Nam. Therefore, M7 is not considered as a realistic alternative to the project activity.
M8	Part of the organic fraction of the solid waste is treated in an anaerobic digester and not disposed in the SWDS	No	Until date, there is no anaerobic digester in Viet Nam. It is not business as usual in Viet Nam. ²⁴ Therefore, M8 is not considered as a realistic alternative to the project activity.
M9	Part of the organic fraction of the solid waste is mechanically or thermally treated to produce RDF/SB and not disposed in the SWDS	No	Until date, MBT technology is only in the stage of pilot in Viet Nam. There was one demonstration project in Thai Nguyen province in 2011, but the project was stopped operating because the main products of RDF and unfired brick was not accepted in the market ²⁵ . It is not business as usual in Viet Nam. Therefore, M9 is not considered as a realistic alternative to the project activity.

²³ Report on the State of the National Environment for the period 2016-2020

²⁴ Report on the State of the National Environment for the period 2016-2020

²⁵ <https://baotainguyenmoitruong.vn/nha-may-xu-ly-va-tai-che-rac-thai-song-cong-kho-thanh-ly-245998.html>

Since the project activity also includes production of electricity generation using combustion heat from incinerators and the export of the electricity to the national grid, then alternative scenarios for the generation of electricity shall also be identified. Alternative(s) shall include:

- (a) P1: Electricity generated as an output of one of the waste treatment options listed in Table 1, not undertaken as a VCS project activity;
- (b) P2: Use of an existing or construction of a new on-site or off-site fossil fuel fired cogeneration plant;
- (c) P3: Existing or new construction of an on-site or off-site renewable based cogeneration plant;
- (d) P4: Existing or new construction of an on-site or off-site fossil fuel fired electricity plant;
- (e) P5: Existing or new construction of an on-site or off-site renewable based electricity plant;
- (f) P6: Electricity generation in existing and/or new grid-connected electricity plants.

For component of electricity generation, assessment of potential alternative scenarios is list below:

Description of alternative scenarios		Plausible and credible alternative?	Rationale
P1	Electricity generated as an output of one of the waste treatment options listed in Table 1, not undertaken as a VCS project activity	Yes	There are no mandatory regulatory or requirements to undertake the project activity. Neither do the regulations prevent/prohibit implementation of the project activity. Thus, P1 is considered as a baseline alternative.
P2	Use of an existing or construction of a new on-site or off-site fossil fuel fired cogeneration plant	No	P2 and P3 alternatives involve cogeneration plants with heat and power outputs. However, the project activity only supplies electricity to the grid, the alternatives of P2 and P3 are not considered as the baseline scenarios.
P3	Existing or new construction of an on-site or off-site renewable based cogeneration plant		
P4	Existing or new construction of an on-site or off-site fossil fuel fired electricity plant	No	The project activity is seeking to generate electricity primarily for sale to the national grid, rather than to meet existing energy requirement. Therefore, the project proponent shall not implement an on-site or off-site fossil fuel fired power plant to cater to specific capture or other requirements. Thus, P4 is not considered as a realistic baseline.

P5	Existing or new construction of an on-site or off-site renewable based electricity plant	No	The project activity is seeking to generate electricity primarily for sale to the national grid, rather than to meet existing energy requirement. Therefore, the project proponent shall not implement an on-site or off-site renewable based power plant to cater to specific capture or other requirements. Thus, P5 is not considered as a realistic baseline.
P6	Electricity generation in existing and/or new grid-connected electricity plants	Yes	In the absence of the project activity, no electricity is exported into the grid, consequently the same amount of electricity would have been generated by the grid-connected power plants.

Therefore, outcome of sub-step 1a, the alternative scenarios are remained including:

- **M1:** The project activity without being registered as a VCS project activity (i.e. any (combination) of the waste treatment options listed in Table 2 of methodology)
- **M3:** Disposal of the fresh waste in a SWDS without a LFG capture system
- **P1:** Electricity generated as an output of one of the waste treatment options listed in Table 1, not undertaken as a VCS project activity
- **P6:** Electricity generation in existing and/or new grid-connected electricity plants

The possible combinations of baseline are as follows:

MSW treatment Electricity generation	M1	M3
P1	Option 1: The combination is the proposed project activity without VCS registration.	The combination is not applicable because the disposal of the fresh waste in a SWDS without a LFG capture system, then there is no LFG for electricity generation.
P6	The combination is not applicable, because the proposed project activity is to utilize MSW to generate electricity and replaced the electricity from the grid.	Option 2: The combination is possible. The disposal of the fresh waste in a SWDS without a LFG capture system and the electricity generation in existing and/or new grid-connected electricity plants are business as usual in Vietnam, it is considered as the baseline scenario.

Therefore, 2 combination options of baseline including Option 1 (M1+P1) and Option 2 (M3 + M6) are considered further.

Sub-step 1b: Consistency with mandatory laws and regulations:

For component of solid waste treatment, legal framework on waste management was developed in Viet Nam with relevant laws and regulations as follows:

- Law on Environmental Protection No 72/2020/QH14, dated on 17/11/2020.
- Decision No. 491/QĐ-TTg dated on 07/05/2018 of the Prime Minister on approving adjustments to national strategy for general management of solid waste to 2025 with vision towards 2050
- Decree No. 38/2015/ ND-CP dated 24/04/2015 of the Government on management of waste and scraps

Decree No. 38/2015/ ND-CP states that the solid waste treatment technology shall be chosen in accordance with available technology feasibility and local conditions. The solid waste treatment technologies can be selected such as composting; incineration; sanitary landfill; technology of recycle, energy recovery, product producing; advanced and other environmentally-friendly technologies. Therefore, all above alternatives (M1, M3) are consistent with mandatory laws and regulations in Viet Nam.

For component of electricity generation, the alternative P1 is consistent with mandatory laws and regulations of Viet Nam as it was issued the Investment Certificate by People's Committee of Hanoi. While P6 is a business-as-usual scenario in electricity generation.

Step 2: Barrier analysis

Step 2a: Identify barriers that would prevent the implementation of alternative scenarios

For the Option 1 (M1 + P1), no barrier other than insufficient financial returns as analyzed in Step 3 is identified. In terms of Combination Option 2 (M3+P6), it doesn't face any barrier as it is the current practice.

Step 2b: Eliminate alternative scenarios which are prevented by the identified barriers

From step 2a, there is no barrier other than insufficient financial returns as analyzed in Step 3. And there is two remaining alternatives to the project activity, which is outcome of step 2a, thus the proposed project activity is not the first-of-its-kind, then proceed to Step 3.

Step 3: Investment analysis

Determine whether the proposed project activity is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of emission reductions

To conduct the investment analysis, the following sub-steps are applied:

Sub-step 3a: Determine appropriate analysis method

Tool07 – “Tool for the demonstration and assessment of additionality”, Version 7.0.0 provides three methods of investment analysis: simple cost analysis (Option I), investment comparison analysis (Option II) and benchmark analysis (Option III).

The proposed project activity generates financial and economic benefits from treatment of MSW and sale of the generated electricity other than VCU revenues, so the simple cost analysis (Option I) is not applicable. Out of the two remaining options, Option II is also not applicable as there are no other credible and realistic baseline scenario alternatives other than Combination of Disposal of fresh waste in a landfill without a LFG capture system and Electricity supply from the grid. Thus, the benchmark analysis (Option III) is chosen to prove additionality.

Sub-step 3b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen. In the following, Project IRR is used to demonstrate the additionality of the project.

As indicated in paragraph 15, Tool27 – “Investment analysis”, Version 11.0, “Local commercial lending rates or weighted average costs of capital (WACC) are appropriate benchmarks for a project IRR”, the project participant applies the local commercial lending rates as a benchmark

for this project IRR. This benchmark is derived from the average long-term lending rates available from the beginning of calculated year up to the date of decision making. All data is sourced from weekly reports published by the State Bank of Vietnam on its official website (<https://www.sbv.gov.vn/en>).

The benchmark is 11% at the date of making the investment decision on 16 December 2018.

Sub-step 3c: Calculation and comparison of financial indicators

The key assumptions used to calculate the Project IRR of the proposed project are presented in the table below:

Table 5. Key assumptions for investment analysis of Soc Son waste to power project plant

No	Parameter	Unit	Value	Source
1	MSW treatment			
1.1	MSW treatment capacity	Tons/year	1,460,000	Feasibility Study Report (FSR) of the project. MSW treatment capacity of the project is estimated about 4,000 ton/day and 365 operating days per year
1.2	Tipping fee	USD/ton	21	Contract No. 08/2018/HĐVXLCTSH on MSW treatment and electricity generation dated 09 Aug 2018
2	Electricity			
2.1	Annual net electricity generation	GWh	388.05	FSR
2.2	Electricity price	USCent/kWh	10.05	Decision No. 31/2014/QĐ-TTg on supporting mechanism for development of power generation projects using solid waste in Viet Nam.
3	Brick production			
3.1	Number of brick	Bricks/year	96,263,736	EIA
3.2	Unit price of brick	USD/brick	0.037	EIA
4	Total investment cost	USD	382,364,103	FSR
5	Operation cost			
5.1	Annual O&M cost	USD/year		

	Year 1 – 10 th (rate of equipment and construction costs)	USD/year	7.5%	FSR
	Year 11 th – 20 th (increase compared with first year)	USD/year	20%	FSR
	Year 20 th – 25 th (increase compared with the first year)	USD/year	30%	FSR
5.2	Fly ash treatment cost	USD/year	846,429.90	FSR
5.3	Fuel, chemical cost	USD/year	18	FSR
5.4	Cost for brick production	USD/year	2,847,611	EIA and calculated in “Soc Son WtE IRR and ER Spreadsheet”, version 3.0
6	Preparation and construction period	year	3	FSR
7	Period of financial assessment (Life time of the project)	year	25	FSR Tool to determine the remaining lifetime of equipment, EB 50, Annex 15/ boiler, steam turbines
8	Residual/ salvage value	USD	0	According to Paragraph 6 of Tool27 - “Investment Analysis”, Version 11.0, “IRR calculations should reflect the period of expected operation of the underlying project activity (technical lifetime) and if a shorter period than the technical lifetime is chosen, the investment analysis shall be conducted for at least 10 years and include fair value of the project activity assets at the end of the assessment period”. Paragraph 7 of Tool27 - “Investment Analysis”, Version 11.0, “The fair value should be calculated in accordance with local accounting regulations

				where available, or international best practice". Since the maximum depreciation period for waste to power projects according to Circular No.45/2013/TT-BTC issued by Ministry of Finance dated 25/04/2013 regarding "Guiding regulation on management, use and depreciation of fixed assets" is 20 years, which is smaller than the financial analysis period of 25 years for this project activity. Furthermore, the technical lifetime of 25 years has been applied as the financial analysis period, full value of assets has been completely depreciated thus no residual/ fair value remains at the end of the assessment period.
9	Pre-tax Project IRR	%	9.25	Calculated in "Soc Son WtE IRR and ER Spreadsheet", version 3.0

The above shows that Project IRR of the proposed project was lower than the benchmark at the time of decision making which is defined as the date of issuing the decision to invest the project by the Management Board on 16/12/2018.

Sub-step 3d: Sensitivity Analysis

A sensitivity analysis of the project activity has been conducted to test the robustness of the above calculations. Although the O&M cost accounted less than 20% of total investment cost that could be disregarded in the sensitivity analysis as regulated under the guidance in Tool27 – "Investment analysis" – Version 11.0, paragraph 27, which states that "variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenue should be subjected to reasonable variation", this parameter is still considered in the analysis below. The following parameters are used in the sensitivity analysis of the project activity:

- Electricity output
- Electricity price
- Total investment cost
- O & M cost

- Tipping fee
- Quantity of waste

Table below shows the impact of variations in key factors on the Project IRR considering a $\pm 10\%$ variation in the parameters.

Table 6. Sensitivity analysis for the plant

No	Parameter	Variation	Project IRR	Likelihoods to happen
1	Electricity output and exported to national grid	+18.19%	11%	The probability of a 18.19 % increase in annual export to the national grid is very unlikely. Electricity output is mainly influenced by: a) quantity of MSW incinerated; b) the average MSW calorific value; c) Gross electricity efficiency and the plant consumption rate: a) Annual maximum volume of treatment capacity of MSW is established strictly as per the specifications of the equipment and the treatment capacity of the incinerator units (800 tons/day each unit *5 units), it is not technically feasible to increase the annual MSW volume upward of maximum designed standard by 18.19%. b) Average MSW calorific value: The caloric value of waste stream in Viet Nam is low, only from 900-1,100 kcal/kg (3,762 – 4,598 kJ/kg)²⁶. While the FSR adopted caloric value of 7,120 kJ/kg for design the system (boiler – generator). Therefore, increase of MSW's calorific value by 18.19% is unlikely occurred. c) Gross electricity efficiency and the electricity consumption rate are estimated consistent with the equipment specifications and the designing system, which are unlikely to be fluctuated significantly. Therefore, it is reasonable to assume that a significant upward by 18.19% each year for the annual electricity supplied to the grid is not possible. In conclusion, it is unlikely that the electricity output and exported to the

²⁶ http://dwrn.gov.vn/uploads/news/2020_11/bao-cau-hien-trang-moi-truong-2019.pdf

				national grid is increased by 18.19%. This option shall be discarded.
		+10%	10.23%	Lower than the benchmark of 11%
		-10%	8.23%	Lower than the benchmark of 11%
2	Total investment cost	+10%	8.20%	Lower than the benchmark of 11%
		-10%	10.48%	Lower than the benchmark of 11%
		-13.76%	11%	The probability of a 13.76% decrease in the total investment cost is not likely to happen. Total investment cost is estimated by the third party, which is mentioned in the FSR and approved by Ministry of Construction. As per signed Equipment purchase contract, equipment cost was the same to the estimated in FSR. Furthermore, According to the General Statistics Office of Viet Nam, the Consumer Price Index (CPI) of housing and construction materials in Viet Nam show 102.50%; 104.27%; 103.32%; 103.03%, 101.79% in 2016, 2017, 2018, 2019 and 2020 respectively, which always increase compared to the previous year. While FSR was finalized in 10/2018 and expected to operate on Jan 2022, it is highly unlikely that the investment cost is decreased by 13.76%, and this option shall be discard. (Source: https://www.gso.gov.vn/)
3	O & M cost	+10%	8.59%	Lower than the benchmark of 11%
		-10%	9.89%	Lower than the benchmark of 11%
		-28.28%	11%	The probability of a -28.28% decrease in O&M cost to meet the benchmark is not possible to happen. According to General Statistics Office of Viet Nam, The Consumer Price Index (CPI) show an annual increase of 104.09%; 100.63%; 102.66%; 103.53%; 103.54%; 102.79%; 103.23% in 2014, 2015, 2016, 2017, 2018, 2019, and 2020 respectively in Viet Nam compared to the previous year. The inflation can be forecasted in the near future. The probability of such a

				decrease in operational cost is very small. Therefore, it is unlikely that the annual O&M cost is decreased by -28.28%. This option shall be discarded. (Source: https://www.gso.gov.vn/)
4	Electricity price	+18.19%	11%	The probability of a 18.19% increase in feed in tariff annually is very unlikely because the PPA was signed with EVN with a fixed feed in tariff for long term. And in fact, the signed price in the PPA is 10.05 UScent/kWh and fixed for 20 years. The increase of electricity tariff for only 5 years (from 21st year) if any is not feasible to reach the benchmark. Moreover, the electricity tariff of the project is a tariff already regulated by government under which is the first decision on supporting mechanism for development of power generation projects using solid waste in Viet Nam. And it is unlikely to be increased greatly. Refer to FIT for solar projects, it found that there is a decrease trend of FIT. In the first Decision No.11/2017/QĐ-TTg on promotion mechanism on solar projects in Vietnam, the FIT1 is 9.35 UScent/MWh for grid-connected solar power projects with COD before 30/06/2019. For Decision 13/2020/QĐ-TTg on Mechanisms to Promote the Development of Solar Power Projects in Viet Nam which is replaced from Decision No.11/2017/QĐ-TTg., the FIT2 are 7.69 UScent/MWh; 7.09 UScent/MWh; 8.36 UScent/MWh for floating solar power project, ground-mounted solar and rooftop solar respectively for COD before 31/12/2020. It is found that there is a decrease trend of FIT for renewable projects as per regulated by Government. Thus, this option shall be discarded.
		+10%	10.23%	Lower than the benchmark of 11%
		-10%	8.23%	Lower than the benchmark of 11%

5	Tipping fee	+ 23.13%	11%	The probability of a 23.13 % increase in tipping fee is very unlikely because the PO signed the MSW treatment contract with the tipping fee of 21 USD/ton. The value is also larger than in Can Tho waste to energy plant, which has operated well treatment of MSW in Viet Nam. The tipping fee can be only reviewed and adjusted in case of there are much fluctuation in price of relevant materials, fuels, equipment or the change in environmental legislation. Moreover, this tipping fee is also higher than in Can Tho waste to energy plant, which has solely operated treatment of MSW in Viet Nam (362,000 VND²⁷ or equivalent 15.7 USD). Thus, tipping fee is unlikely to fluctuate dramatically by +23.13% during the operation year.
		+10%	10.03%	Lower than the benchmark of 11%
		-10%	8.45%	Lower than the benchmark of 11%
6	Quantity of waste	+23.13%	11%	The probability of a 23.13% increase in quantity of waste is not likely to happen. The treatment capacity of the project is limited as mentioned in MSW treatment contract. Annual maximum volume of disposal capacity of MSW is established strictly as per the specifications of the equipment and the treatment capacity of the incinerator units. Moreover, as per financial calculation, operating time is for 365 days, thus it is not technically feasible to increase quantity of waste received to 23.13%.
		+10%	10.03%	Lower than the benchmark of 11%
		-10%	8.45%	Lower than the benchmark of 11%

The result of the project sensitivity analysis is shown in Figure below:

²⁷ <https://sqsvn/bo-tnmt-khao-sat-nha-may-dot-rac-sinh-hoat-phat-dien-can-tho/>

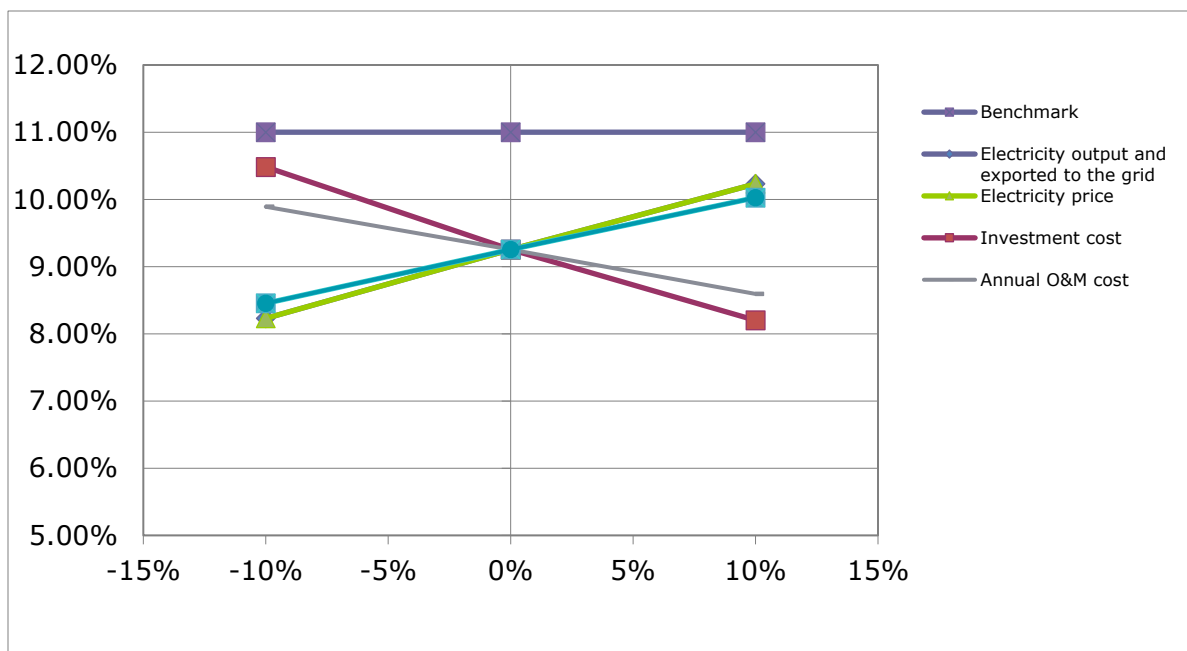


Figure 4: Project sensitivity analysis

The sensitivity analysis shows that there is unlikely to be happened any case in which the variation of a parameter can make the project IRR without VCU revenues reach the benchmark.

In conclusion, the proposed VCS project activity or Option 1 (Combination of Alternative M1 and P1) is unlikely to be financially attractive. Therefore, Option 2 (combination of Alternatives M3 and P6) are excluded from further consideration as a baseline scenario.

3.5 Additionality

Since the tool02 “Combined tool to identify the baseline scenario and demonstrate additionality”, version 07.0 is applied to the proposed project activity, step 1 to step 3 was conducted in section 3.4 Baseline scenario, remaining following steps continue to be implemented for demonstration of additionality.

Step 4: Common practice analysis

Sub-step 4a: The proposed project activity(s) applies measure(s) that are listed in the definitions section above

Stepwise approach for common practice analysis has been carried out as per Methodological Tool24 - “Common Practice”, Version 03.1.

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

The proposed project activity has the installed capacity of 4,000 tons/day (approximately 1,460,000 tons/year). So applicable capacity range as +/-50% is from 2,000 tons/day to 6,000 tons/day.

Step (2): Identify similar projects (both CDM/VCS and non-CDM/VCS) which fulfil all of the following conditions:

- a) The projects are located in the applicable geographical area;
- b) The projects apply the same measure as the proposed project activity;
- c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;
- e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;
- f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM/VCS and non-CDM/VCS) is carried out as per sub-steps of Step (2) as follows:

- a) The proposed project is located in Viet Nam so the entire host country (Viet Nam) was chosen as the applicable geographical area.
- b) The proposed project activity is a Greenfield waste to power plant. Therefore, all the MSW treatment plants and generating electricity (the same technology as the proposed project activity) are candidates for similar projects.
- c) The energy source used by the proposed project activity is MSW. Hence, only waste to power plants are candidates for similar projects.
- d) The proposed project activity produces electricity from MSW. Therefore, all the waste to power plants are candidates for similar projects.
- e) The capacity range of the similar projects is within the applicable capacity range from 2,000 tons/day to 6,000 tons/day.
- f) Since CDM Tools/Guidelines are followed for common practice analysis, the start date needs to be considered as per CDM guidelines for common practice analysis. The start date of the project under the Methodological tool for common practice is the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system). Where a contract is signed for such expenditures (e.g. for procurement of a wind turbine), it is the date on which the contract is signed. In other cases, it is the date on which such expenditures are incurred. If the project activity involves more than one of such contracts or incurred expenditures, it is the first of the respective dates. Activities incurring minor pre-project expenses (e.g. feasibility studies, preliminary surveys) are not considered in the determination of the start date.

The equipment purchase contract between Ha Noi Thien Y Environmental Energy JSC and Jiangsu Tianying Environmental Energy Complete Equipment Co., Ltd was signed on 18/12/2018, is the first contract for expenditures of the proposed project activity and it is considered as the start date of the project activity for common practice analysis. The waste to power projects, which have started commercial operation before 18/12/2018 have been considered for analysis.

Table 7. List of solid waste to energy projects in Viet Nam²⁸

No	Name of projects/ Location	Investor	Type of solid waste	Capacity (tons of MSW/day)	Commission year	Developed as CDM project?
1	Nam Son Waste to Energy Project, Ha Noi	State owner (URENCO Ha Noi) Funded by NEDO	Industrial waste ²⁹	4,000	12/2016	No
2	Can Tho converting domestic solid wastes to produce electricity, Can Tho	Everbright International	MSW	400	12/2018 ³⁰	No

As the above table, there is only two waste to energy projects in Viet Nam which has been operated before 12/2018. Furthermore, project 1 involves treatment of industrial waste instead of MSW and both project 1 and project 2 are not within considered capacity range from 2,000 to 6,000 ton/day.

Thus, numbers of similar projects identified is zero.

Step (3): Within the projects identified in Step (2), identify those that are neither registered CDM/VCS project activity, project activity submitted for registration, nor project activity undergoing validation. Note their number N_{all}.

No similar projects are identified in Step (2), therefore, **N_{all} = 0**

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.

Since numbers of similar project identified is zero, **N_{diff} = 0**.

Step (5): Calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Calculate factor: $F = 1 - N_{diff}/N_{all} = 0$

$$N_{all} - N_{diff} = 0$$

²⁸ https://sustainabledevelopment.un.org/content/unosd/documents/37492.VIETNAM_CountryPresentation.pdf

²⁹ <http://gizenergy.org.vn/vn/article/vn-hanh-nha-may-s-dng-cong-ngh-djin-rac-dju-tien-c-vit-nam>

³⁰ <https://www.moit.gov.vn/web/guest/tin-chi-tiet/-/chi-tiet/nha-may-%C4%91ot-rac-phat-%C4%91ien-can-tho-%C4%91uoc-xac-nhan-hoan-thanh-cong-trinh-bao-ve-moi-truong-20422-129.html>

As per methodological tool “Common practice” - Version 03.1, the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3.

The above conditions are not fulfilled, $F = 0$ is smaller than 0.2 and $N_{all} - N_{diff} = 0$ is smaller than 3. Therefore, the project activity is not a “common practice” within a sector in the applicable geographical area, according to the guideline.

Sub-step 4b: The proposed CDM project activity(s) does not apply any of the measures that are listed in the definitions section above

The project incinerates the MSW and use the heat to produce electricity. And therefore step 4b is not applicable for the project.

In conclusion, the proposed project activity is additional.

3.6 Methodology Deviations

The approved CDM methodology ACM0022 – Alternative waste treatment processes - Version 03.0 has been applied without any deviation. Therefore, this section is not applicable.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per methodology ACM0022, version 03.0, baseline emission is estimated as below equation:

$$BE_y = \sum_t (BE_{CH_4,y} + BE_{WW,y} + BE_{EN,y} + BE_{NG,y}) \times (1 - RATE_{compliance,t}) \text{ [Equation 1 of ACM0002]}$$

Where:

BE_y	Baseline emissions in year y (t CO ₂ e)
$BE_{CH_4,y}$	Baseline emissions of methane from the solid waste disposal site (SWDS) in year y (t CO ₂ e)
$BE_{WW,y}$	Baseline methane emissions from anaerobic treatment of the wastewater in open anaerobic lagoons or of sludge in sludge pits in the absence of the project activity in year y (t CO ₂ e)
$BE_{EN,t,y}$	Baseline emissions associated with energy generation in year y (tCO ₂ e)
$BE_{NG,y}$	Baseline emissions associated with natural gas use in year y (tCO ₂ e)

$RATE_{compliance,t,y}$ Discount factor to account for the rate of compliance of a regulatory requirement that mandates the use of alternative waste treatment process t

The mitigation action does not involve wastewater treatment, or natural gas use in baseline; therefore, $BE_{NG,t,y}$, $BE_{WW,y}$ are considered as zero.

There is currently no regulation on requirement that mandates the use of alternative waste treatment option in Vietnam, thus $RATE_{compliance} = 0$.

Consequently, baseline emission can be simplified as follows:

$$BE_y = BE_{CH_4,y} + BE_{EN,y}$$

Calculation baseline emissions of methane from the SWDS ($BE_{CH_4,y}$)

Baseline emissions of methane from the SWDS are determined using the methodological tool04 “Emissions from solid waste disposal sites”, version 08.0 ($BE_{CH_4,SWDS,y}$)

$$BE_{CH_4,SWDS} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j W_{j,x} \times DOC_j \times e^{-k_j(y-x)} \times (1 - e^{-k_j})$$

[Equation 1 of tool04]

φ_y	Model correction factor to account for model uncertainties for year y
f_y	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
OX	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	Fraction of methane in the SWDS gas (volume fraction)
$DOC_{f,y}$	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
MCF_y	Methane correction factor for year y
$W_{j,x}$	Amount of organic waste type j disposed or prevented from disposal in the SWDS in the year x (t)
DOC_j	Fraction of degradable organic carbon (by weight) in the waste type j (weight fraction)
k_j	Decay rate for the waste type j (1/yr)
j	Type of residual waste or types of waste in the MSW
x	Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period ($x=1$) to year y ($x=y$)
y	Years of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)

No regulation requirements specify the amount or percentage of LFG that should be destroyed or require the installation of a capture system. In fact, a few cells of Nam Son landfill were equipped a LFG capture system for power generation under CDM project. The project however is currently abandoned due to the downturn of the price for CERs. And therefore $f_y = 0$.

Where different waste type j are prevented from disposal, determine the amount of different waste type ($W_{j,x}$) through sampling and calculate the mean from the samples, as follow:

$$W_{j,x} = W_x \times \frac{\sum_{n=1}^z P_{n,j,y}}{z} \quad [\text{Equation 5 and Equation 7 of tool04}]$$

tool04]

Where

- $W_{j,x}$ Amount of solid waste type j disposed or prevent from disposal at SWDS in year x (t)
- W_x Total amount of waste prevented from disposal in year x (t)
- $P_{n,j,y}$ Weight fraction of the waste type j in the sample n collected during the year x (weight fraction)
- z Number of samples collected
- i Types of solid waste
- x Years in the time period for which waste is disposed at the SWDS, extending from the first year in the time period ($x = 1$) to year y ($x = y$).

Since few default values are selected depend on climate condition of the project site, climate conditions in Hanoi is required to be considered. With average temperature (MAT) and rainfall (MAP) are 25.91°C and 1,311.4 mm ³¹ respectively, climate conditions in Hanoi falls in tropical climate, MAP > 1000 mm and wet condition.

Determination of $\varphi_y \times (1 - f_y) \times GWP_{CH_4} \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y$

$\varphi_y \times (1 - f_y) \times GWP_{CH_4} \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y =$		6.375
φ_y	Tool04, page 13 of 25, Application B, default value for wet conditions	0.85
f_y	No methane captured at the baseline SWDS	0
OX	Tool04, page 14 of 25, Default value	0.1
F	Tool04, page 14 of 25, Default value	0.5

³¹ GSO, 2019

DOC _{f,y}	Tool04, page 14 of 25, Application B, Default value	0.5
MCF _y	Tool04, Page 15 of 25, anaerobic managed solid waste disposal sites	1

The Soc Son waste to power plant would receive approximately 1,460,000 tons/year with component solid waste detailed in table below:

Table 8. Components of solid waste

Type of waste (j)	Fraction (p _{j,x})
Wood and wood products	0.0%
Pulp, paper and cardboard	2.7%
Food, food waste, beverages and tobacco	51.9%
Textiles	1.6%
Garden, yard and park waste	0.0%
Glass, plastic, metal, other inert waste	43.8%

The baseline emissions of methane from the SWDS (BE_{CH₄,y}) is summed-up as follow:

Table 9. Baseline emissions of methane from SWDS

Year	Baseline emission of methane BE _{CH₄,y}
	tCO ₂ e
24/07/2022 – 23/07/2023	277,866
24/07/2023 – 23/07/2024	466,829
24/07/2024 – 23/07/2025	596,016
24/07/2025 – 23/07/2026	684,963
24/07/2026 – 23/07/2027	746,778
24/07/2027 – 23/07/2028	790,258
24/07/2028 – 23/07/2029	821,308

Baseline emissions from electricity generation (BE_{EN,y})

$$BE_{EN,y} = BE_{EC,y} + BE_{HG,y} \quad [\text{Equation 13 of ACM0002}]$$

Where

BE_{EN,y} Baseline emissions associated with energy generation in year y (t CO₂)

BE_{EC,y} Baseline emissions associated with electricity generation in year y (t CO₂)

BE_{HG,y} Baseline emissions associated with heat generation in year y (t CO₂)

The mitigation action does not involve heat generation in baseline; therefore, $BE_{HG,y}$ is considered as zero.

The baseline emissions associated with electricity generation in year y ($BE_{EC,y}$) shall be calculated using the tool 05 “Tool to calculate baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0.

$$BE_{EC,y} = \sum_k EC_{BL,k,y} \times EF_{EF,k,y} \times (1 + TDL_{k,y}) \quad [\text{Equation 2 of tool05}]$$

Which:

$EC_{BL,k,y}$	Quantity of electricity generated and exported to the grid by the project in year y (MWh/year)
$EF_{EF,k,y}$	Emission factor of electricity generation for source k in year (tCO ₂ /MWh)
$TDL_{k,y}$	Average technical transmission and distribution losses for providing electricity
k	Sources of electricity consumption in the baseline

The most plausible baseline scenario for electricity generation in the project is “Electricity generation in existing and/or new grid-connected electricity plants”, thus, k means the Viet Nam national grid.

Determination of the emission factor for electricity generation ($EF_{EF,k,y}$)

The determination of the emission factors for electricity generation ($EF_{EF,k,y}$) depends on which scenario (A, B or C) applies to the source of electricity consumption.

For the project, it is scenario A. And Option A1 is chosen for determining $EF_{EF,k,y}$, thus:

$$EF_{EF,k,y} = EF_{grid,CM,y}$$

For this crediting period, the emission factor is calculated using:

- Latest national data: The published data by the host country DNA of Viet Nam, Department of Climate Change, Ministry of Natural Resources and Environment of Viet Nam on 26/02/2021; and
- The Version 07.0 of “Tool to calculate the emission factor for an electricity system” valid from 31/08/2018 onward.

As per version 07.0 of “Tool to calculate the emission factor for an electricity system”, following steps are applied in the calculation of the emission factor for the baseline scenario:

- Step 1:** Identify the relevant electricity systems;
- Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- Step 3:** Select a method to determine the operating margin (OM);
- Step 4:** Calculate the operating margin emission factor according to the selected method;
- Step 5:** Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor

Step 1. Identify the relevant electricity systems

This project will be connected to the national electricity grid of Vietnam, which is operated and monopolized by the EVN. This national electricity grid is the unique transmission and distribution line, to which all power plants in Vietnam are physically connected. Hence the national electricity grid is the project electricity system.

There are electricity imports to the national electricity grid from China - another host country, thus the China Power Grid is the connected electricity system and the emission factor for the imported electricity is zero tons CO₂ per MWh by default.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional)

There are two options to calculate the operating margin and build margin emission factor:

Option 1: Only grid power plants are included in the calculation;

Option 2: Both grid power plants and off-grid power plants are included in the calculation

Because only the data of grid connected power plants is available, so Option 1 will be chosen for calculating the grid emission factor.

Step 3. Select a method to determine the operating margin (OM);

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

- a) Simple OM; or
- b) Simple adjusted OM; or
- c) Dispatch data analysis OM; or
- d) Average OM.

The method (a) "Simple OM" can be chosen in Vietnam because total electricity generation of low cost/must-run resources is less than 50% of total grid generation in average of the five most recent years (detail see in the table below):

Table 10: Rate of low cost/must-run resources based on generation³²

Year	2015	2016	2017	2018	2019	Average value for 2015-2019
Hydro power (MWh)	47,069,435	50,254,951	71,056,945	69,485,682	54,411,106	92,278,120
Bagasse (MWh)	69,500	64,000	78,000	456,400	280,996	948,896
Wind power (MWh)	-	-	-	-	721,189	721,189

³² <http://vie.jcmvietnam.vn/quy-dinh-va-huong-dan/he-so-phat-thai-luoi-dien-viet-nam-nam-2017-a297.html>

Solar power (MWh)	-	-	-	-	4,833,674	4,833,674
Imported power (MWh)	2,393,000	2,736,000	2,361,000	3,124,000	3,316,000	3,930,000
Total power (MWh)	146,014,346	159,817,731	169,942,517	188,063,484	210,596,694	74,434,772
Low-cost/Must-run ratio						35.76%

The data vintage which is used to calculation the Simple Simple OM emission factor is the Ex-ante option of a 3-year generation-weighted average (2017, 2018 and 2019) that is the most recent data available at the time of submission of the VCS-PD to the DOE for validation, without requirement to monitor and recalculate the emissions factor during the crediting period.

Step 4. Calculate the OM emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants units.

Simple OM may be calculated by one of the following two options:

- Option A:** Based on the net electricity generation and a CO₂ emission factor of each power unit; or
- Option B:** Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system

Because the necessary data for Option A is available so Option A “Calculation based on average efficiency and electricity generation of each plant” is used and then the simple OM emission factor is calculated as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

- $EF_{grid,OMsimple,y}$ = Simple operating margin CO₂ emission factor in year y (t CO₂/MWh)
- $EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
- $EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (t CO₂/MWh)
- m = All power units serving the grid in year y except low-cost/must-run power units
- y = Either the three most recent years for which data is available at the time of submission of the VCS-PD to the DOE for validation (ex ante option)

Because the data on fuel consumption and electricity generation of power unit m is available, so the emission factor ($EF_{EL,m,y}$) should be determined as Option A1 :

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_{m,y}}$$

Where:

$EF_{EL,m,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen in Step 3

Table 11: OM emission factor in 2019

Year	Total output (MWh)	Total emission (tCO ₂ e)	$EF_{grid,OM} 2019$ (tCO ₂ /MWh)
	A	B	($\Sigma B / \Sigma A$)
2017	96,446,572.77	83,160,505.77	0.8907
2018	114,997,401.74	106,748,579.11	
2019	143,651,728.86	126,339,447.40	
Total	355,095,703.37	316,270,717.36	

Therefore $EF_{grid,OM,Simple,y}$ is derived as follows:

$$EF_{grid,OM,y} = 0.8907 \text{ tCO}_2/\text{MWh}$$

Step 5. Calculate the BM emission factor

In terms of vintage of data, one of the following two options can be chosen:

Option 1 - For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period, or

Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The most recent information on units already built for sample group m is available, so Option 1 shall be chosen for the proposed project.

The sample group of power units m used to determine as per the following procedure, consistent with the data vintage selected above:

- a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) and determine their annual electricity generation ($AEG_{SET-5 \text{ units}}$, in MWh);

In 2019, the set of five power units that have been built most recently ($SET_{5 \text{ units}}$) is indicated in Appendix 1 has annual generation ($AGE_{SET-5 \text{ units}}$) of 1,128,094.85 MWh.

- b) Determine the annual electricity generation of the project electricity system, excluding power units registered as CDM project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} (if 20 per cent falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20 \text{ per cent}}$) and determine their annual electricity generation ($AEG_{SET \geq 20 \text{ per cent}}$, in MWh)

The total output of Vietnam electricity grid (AEG_{total}) in 2019 is 210,530,694.38 MWh then 20% of the total output of Vietnam electricity grid in 2019 is 42,106,138.876 MWh.

Most recent-built power plants ($SET_{\geq 20\%}$) addition in the electricity system that comprise 20% of the system generation in 2019 is shown in the Appendix 1 have annual electricity generation ($AEG_{SET \geq 20\%}$) of 42,106,138.88 MWh.

- c) From $SET_{5 \text{ units}}$ and $SET_{\geq 20 \text{ per cent}}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample})

The comparison carried out by the project participants shows that the set of power capacity additions in the electricity system that comprise 20% of the system generation (in MWh) that have been built most recently has the larger annual generation (42,106,138.88 MWh) than the set of five power units that have been built most recently in 2019 does (1,128,094.85 MWh), and hence it is employed and SET_{sample} .

There is no plant in SET_{sample} is started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin.

The BM emissions factor is the generation-weighted average emission factor (tCO_2 / MWh) of all power units m during the most recent year y for which power generation data is available. It is calculated as follows:

$$EF_{grid, BM, y} = \frac{\sum_m EG_{m, y} \times EF_{EL, m, y}}{\sum_m EG_{m, y}}$$

Where:

$EF_{grid, BM, y}$	=	Build margin CO_2 emission factor in year y (tCO_2 / MWh)
$EG_{m, y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL, m, y}$	=	CO_2 emission factor of power unit m in year y ($t CO_2 / MWh$)
m	=	Power units included in the build margin
y	=	Most recent historical year for which power generation data is available

Table 12: OM emission factor in 2017

Total output	EG 2019 (MWh)	42,106,138.88	Comprising 20.10% of the system generation in 2019
Total emission	EG*EF 2019 (tCO ₂)	33,886,270.57	
EF_{grid,BM,y}	EF _{BM} (tCO ₂ /MWh)	0.8010	

Then, EF_{grid,BM,y} is derived as follows:

$$EF_{grid,BM,y} = 0.8010 \text{ tCO}_2/\text{MWh}$$

Step 6. Calculate the CM emission factor

According to the tool, the calculation of the combined margin (CM) emission factor (EF_{grid,CM,y}) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM

As the project is located in Vietnam a developing country and having more than 10 registered projects at starting date of validation, the PP chooses the weighted average CM method to calculate CM emission factor for the proposed project.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

EF_{grid,CM,y} = Combined margin CO₂ emission factor of national grid in year y (tCO₂/MWh)

EF_{grid,OM,y} = Operating margin CO₂ emission factor of national grid in year y (tCO₂/MWh)

W_{OM} = Weighting of operating margin emissions factor (%)

EF_{grid,BM,y} = Build margin CO₂ emission factor of national grid in year y (tCO₂/MWh)

W_{BM} = Weighting of build margin emissions factor (%)

The DNA report applied the weighting factors of both OM and BM emission are 0.5, which is applicable for the project activity in the first crediting period and subsequent crediting periods.

So the CM emissions factor is derived as follows:

$$EF_{grid,CM,y} = 0.5 \times 0.8907 + 0.5 \times 0.8010 = 0.8458 \text{ tCO}_2/\text{MWh}$$

As per above equation, baseline emissions from electricity generation is estimated as follow:

$BE_{EC,y} = EC_{BL,k,y} \times EF_{EF,k,y} \times (1 + TDL_{k,y}) =$		338,059 tCO₂
$EC_{BL,k,y}$	FSR	388,050 MWh
$EE_{EF,k,y}$	Calculated	0.8458 tCO ₂ /MWh

$TDL_{k,y}$	Default values for baseline electricity consumption sources, Tool05, page 15 of 25	3%
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Finally, baseline emissions are determined as table below:

Year	Baseline emission of methane from SWDS ($BE_{CH_4,y}$)	Baseline emissions from electricity generation ($BE_{EC,y}$)	Baseline emission BE_y
	tCO ₂ e	tCO ₂ e	tCO ₂ e
24/07/2022 – 23/07/2023	277,866	338,059	615,925
24/07/2023 – 23/07/2024	466,829	338,059	804,888
24/07/2024 – 23/07/2025	596,016	338,059	934,075
24/07/2025 – 23/07/2026	684,963	338,059	1,023,022
24/07/2026 – 23/07/2027	746,778	338,059	1,084,837
24/07/2027 – 23/07/2028	790,258	338,059	1,128,317
24/07/2028 – 23/07/2029	821,308	338,059	1,159,367

4.2 Project Emissions

The below equation is used for estimating project emissions of the project activity according to the methodology:

$$PE_y = PE_{COMP,y} + PE_{AD,y} + PE_{GAS,y} + PE_{RDF-SB,y} + PE_{INC,y} \quad [\text{Equation 17 of ACM0022}]$$

Where

PE_y	Project emissions in year y (t CO ₂ e)
$PE_{COMP,y}$	Project emissions from composting or co-composting in year y (t CO ₂ e)
$PE_{AD,y}$	Project emissions from anaerobic digestion and biogas combustion in year y (t CO ₂ e)
$PE_{GAS,y}$	Project emissions from gasification in year y (t CO ₂ e)
$PE_{RDF-SB,y}$	Project emissions associated with RDF/SB in year y (t CO ₂ e)
$PE_{INC,y}$	Project emissions from incineration in year y (t CO ₂ e)

Since the project activity doesn't involve composting or co-composting, anaerobic digestion and biogas combustion, gasification, RDF/SB; project emissions associated with these are considered as zero. The above equation is simplified as follows:

$$PE_y = PE_{INC,y}$$

Project emissions are therefore determined as follows:

$$PE_y = PE_{INC,y} = PE_{COM,INC,y} + PE_{EC,INC,y} + PE_{FC,INC,y} + PE_{ww,INC,y} \text{ [Equation 20 of ACM0022]}$$

PE_y	Project emissions of the project activity in year y (tCO ₂ e)
$PE_{COM,y}$	Project emissions from combustion within the project boundary of fossil waste associated with incineration in year y (t CO ₂)
$PE_{EC,y}$	Project emissions from electricity consumption in year y (tCO ₂ e)
$PE_{FC,y}$	Project emissions from fossil fuel consumption (tCO ₂ e)
$PE_{ww,y}$	Project emissions from the wastewater treatment (tCO ₂ e)

Project emission from electricity consumption ($PE_{EC,INC,y}$)

The project emissions associated with electricity generation in year y ($PE_{EC,y}$) shall be calculated using the tool 05 “Tool to calculate baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0.

$$PE_{EC,y} = \sum_t EC_{t,y} \times EF_{EF,t,y} \times (1 + TDL_{t,y}) \text{ [Equation 1 of tool05]}$$

$EG_{PJ,y}$	Quantity of electricity consumed by the project in year y (excluding consumption of electricity that was generated by the project activity) (MWh/year)
$EF_{EF,t,y}$	Emission factor for electricity generation for source t in year y (tCO ₂ /MWh)
$TDL_{t,y}$	Average technical transmission and distribution losses for providing electricity
t	Sources of electricity consumption in the project

Determination of the emission factor for electricity generation ($EF_{EF,t,y}$)

The determination of the emission factors for electricity generation ($EF_{EF,t,y}$) depends on which scenario (A, B or C) applies to the source of electricity consumption.

For the project, it is scenario A. And Option A1 is chosen for determining $EF_{EF,t,y}$, thus:

$$EF_{EF,t,y} = EF_{grid,CM,y}$$

As calculated above, $EF_{grid,CM,y} = 0.84585$ tCO₂/MWh

The electricity consumed on-site is expected from electricity generated by the project activity, no electricity is imported from the grid or other power plants. Therefore, the project emission is considered as zero.

$$PE_{EC,y} = 0 \text{ (1)}$$

Project emission from fossil fuel consumption ($PE_{FC,y}$)

The project emission is determined as per tool03 “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”, version 03.0 as follow:

$$PE_{FC,y} = \sum_i FC_{i,y} \times COEF_{i,y} \text{ [Equation 1 of tool03]}$$

$FC_{i,y}$	Quantity of fuel type i combusted in process j during the year y (mass or volume unit/year)
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$COEF_{i,y}$ CO2 emission coefficient of fuel type i in year y (tCO₂/mass or volume unit)
i Fuel types combusted

The CO2 emission coefficient, is calculated using Option B in this tool:

$$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i,y} \quad \text{[Equation 4 of tool03]}$$

Where:

$NCV_{i,y}$ Weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)

$EF_{CO2,i,y}$ Weighted average CO2 emission factor of fuel type i in year y (tCO₂/GJ)

The Diesel oil (DO) will be used to start-up the incinerators or be auxiliary for boilers if needed. As per FSR, annually the DO consumed will be 1200 ton/year. Therefore, the project emission from fossil fuel is as below:

(2)

$PE_{FC,y} = FC_{DO,y} \times NCV_{DO,y} \times EF_{CO2,DO,y} =$		3,887 tCO ₂
$FC_{DO,y}$	FSR	1,200 t/year
$NCV_{DO,y}$	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Section 1.4, Upper limit.	43.3 TJ/Gg or 0.0433 GJ/t
$EF_{CO2,DO,y}$	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Section 1.4, Upper limit.	74.8 tCO ₂ /GJ

Project emissions from combustion within the project boundary ($PE_{COM,y}$)

$$PE_{COM,INC,y} = PE_{COM,CO2,y} + PE_{COM,CH4,N2O,y} \quad \text{[Equation 21 of ACM0022]}$$

$PE_{COMP,CO2,y}$ Project emissions of CO2 from combustion within the project boundary associated with combustor in year y (t CO2)

$PE_{COM,CH4,N2O,y}$ Project emissions of CH4 and N2O from combustion within the project boundary associated with combustor c in year y (t CO2)

Project emissions of CO₂ from combustion within the project boundary ($PE_{COMP,CO2,y}$)

Carbon dioxide project emissions associated with on-site combustion ($PE_{COM_CO2,c,y}$) are calculated based either on the fossil carbon content of the fresh waste or RDF/SB combusted, or on the fossil carbon content of the stack gas. The biogenic carbon content is not considered.

Option 1 in the methodology is selected to calculate $PE_{COMP,CO2,y}$. Details are as follows:

Option 1: Waste sorted into waste type fractions

$$PE_{COM,CO2,y} = EFF_{COM,y} \times \frac{44}{12} \times \sum_j Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y} \quad [\text{Equation 22 of ACM0022}]$$

$Q_{j,c,y}$	Quantity of fresh waste type j fed into combustor in year y (t)
$FCC_{j,y}$	Fraction of total carbon content in waste type j in year y (t C/t)
$FFC_{j,y}$	Fraction of fossil carbon in total carbon content of waste type j in year y (weight fraction)
$EFF_{COM,y}$	Combustion efficiency of combustor c in year y (fraction)
$44/12$	Conversion factor (t CO ₂ / tC)
$P_{j,y}$	Fraction of waste type j in during the year y (weight fraction)
j	Waste type

Project Participants select to calculate this parameter based on monitoring the total waste fed to the combustor and sampling the waste to determine the fraction of waste type j as per the following equation:

$$Q_{j,c,y} = Q_{waste,c,y} \times \frac{\sum_{n=1}^z p_{nj,y}}{z} \quad [\text{Equation 23 of ACM0022}]$$

Where

$Q_{j,c,y}$	Quantity of fresh waste type j fed into combustor in year y (t)
$Q_{waste,c,y}$	Quantity of fresh waste or RDF/SB fed into combustor c in year y (t)
$P_{n,j,y}$	Fraction of waste type j in the sample n collected during the year y (weight fraction)
z	Number of samples collected during the year y
n	Samples collected in year y
j	Waste type

Calculation of $\sum_j Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y}$

As per FSR, quantity of fresh waste fed into combustor is estimated as 1,460,000 ton/year (4,000 ton/day). Thus, $\sum_j Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y}$ is determined as follow:

Waste type (j)	Fraction of waste type j ($p_{nj,y}$)	$Q_{j,c,y}$	$FCC_{j,y}$	$FFC_{j,y}$	$Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y}$
	FSR	Calculated	ACM0022, version 03.0, Page 43/81	ACM0022, version 03.0, Page 42/81	Calculated
	-	t	tC/t waste	t C (fossil)/t C	tC (fossil)

Paper and Cardboard	2.7%	39,420	50%	5%	985.50
Textiles	1.6%	23,360	50%	50%	5,840.00
Food waste	51.9%	757,740	50%	0%	0.00
Wood	0.0%	-	54%	0%	0.00
Garden and Park waste	0.0%	-	55%	0%	0.00
Nappies	0.0%	-	90%	10%	0.00
Rubber and leather	1.3%	18,980	67%	20%	2,543.32
Plastic	3.0%	43,800	85%	100%	37,230.00
Metal	0.9%	13,140	0%	0%	0.00
Glass	0.5%	7,300	0%	0%	0.00
Other, inert waste	38.1%	556,260	5%	100%	27,813.00
Total	100.0%	1,460,000			74,411.82

Therefore, project emission of CO₂ from combustion within the project boundary ($PE_{COM,CO_2,y}$) is:

$PE_{COM,CO_2,y} = EFF_{COM,y} \times \frac{44}{12} \times \sum_j Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y} =$		272,843 tCO₂
$\sum_j Q_{j,c,y} \times FCC_{j,y} \times FFC_{j,y}$	Calculated	74,411.82 tC (fossil)
44/12	Conversion factor	t CO ₂ / tC
$EFF_{COM,y}$	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 5, Chapter 5, section 5.4.1.3	100%

Project emissions of CH₄ and N₂O from combustion within the project boundary ($PE_{COM,CH_4,N_2O,y}$)

Project participants choose Option 2 that calculates the emissions using default emission factors for the amount of N₂O and CH₄ emitted per tonne of fresh waste combusted.

Option 2: Using default emission factors

$$PE_{COM,CH_4,N_2O,y} = Q_{waste,y} \times (EF_{N_2O} \times GWP_{N_2O} + EF_{CH_4} \times GWP_{CH_4})$$

[Equation 27 of ACM0022]

$Q_{waste,y}$	Quantity of fresh waste fed into combustor in year y (t)
EF_{N_2O}	Emission factor for N_2O associated with waste treatment option (t N_2O /t waste)
EF_{CH_4}	Emission factor for CH_4 associated with treatment option (t CH_4 /t waste)
GWP_{CH_4}	Global Warming Potential of methane valid for the commitment period (t CO_2e /t CH_4)
GWP_{N_2O}	Global Warming Potential of nitrous oxide (t CO_2e /t N_2O)

Then, project emission of CH_4 , N_2O from combustion within the project boundary ($PE_{COM,CH_4,N_2O,y}$) is estimated as below:

$PE_{COM,CH_4,N_2O,y} = Q_{waste,y} \times (EF_{N_2O} \times GWP_{N_2O} + EF_{CH_4} \times GWP_{CH_4}) =$		23,418 t CO_2
$Q_{waste,y}$	FSR	1.460.000 ton/year
EF_{N_2O}	ACM0022, version 03.0, Page 46 of 81	1.21×10^{-6} t N_2O /twaste
EF_{CH_4}	ACM0022, version 03.0, Page 45 of 81, continuous stoker incinerator	1.21×10^{-6} t CH_4 /twaste
GWP_{CH_4}	IPCC Fifth Assessment report AR5 on Climate Change (2014)	28 t CO_2e /t CH_4
GWP_{N_2O}	IPCC Fifth Assessment report AR5 on Climate Change (2014)	265 t CO_2e /t N_2O

Thus, project emissions from combustion within the project boundary ($PE_{COM,y}$):

$$PE_{COM,y} = PE_{COM,CO_2,y} + PE_{COM,CH_4,N_2O,y} = 272,843 + 23,418 = 296,261 \text{ t}CO_2e \quad (3)$$

Project emissions from run-off wastewater management ($PE_{ww,t,y}$)

Wastewater (leachate included) from the project activity will be collected and treated anaerobically in UASB, then moved to membrane bio reactor (MBR), Nanofiltration (NF) – Reverse osmosis (RO) Filters to meet Viet Nam Standards before discharging the environment. Therefore, the project emissions from run-off wastewater is calculated using equation:

$$PE_{ww,y} = Q_{ww,y} \times P_{COD,y} \times B_o \times MCF_{ww} \times GWP_{CH_4} \quad [\text{Equation 28 of ACM0022}]$$

$Q_{ww,y}$	Amount of run-off wastewater generated by the project activity and treated anaerobically or released untreated from the project activity in year y (m^3)
$P_{COD,y}$	COD of the run-off wastewater generated by the project activity in year y (tCOD/ m^3)

B_o	Maximum methane producing capacity, expressing the maximum amount of CH ₄ that can be produced from a given quantity of chemical oxygen demand (t CH ₄ /tCOD)
MCF_{ww}	Methane conversion factor (fraction)
GWP_{CH4}	Global Warming Potential of methane valid for the commitment period (t CO ₂ e/t CH ₄)

(4)

$PE_{ww,y} = Q_{ww,y} \times P_{COD,y} \times B_o \times MCF_{ww} \times GWP_{CH4} =$		120,187 tCO₂
$Q_{ww,y}$	EIA, page 248 (1680 m ³ /day *365 days)	613,200 m ³ /year
$P_{COD,y}$	EIA, page 252	35,000 mg/l or 0.035 tCOD/m ³
B_o	ACM0022, version 03.0, page 44 of 81	0.25 tCH ₄ /tCOD
MCF_{ww}	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 5, chapter 6, table 6.3	0.8
GWP_{CH4}	IPCC Fifth Assessment report AR5 on Climate Change (2014)	28 tCO ₂ e/t CH ₄

Finally, from (1), (2), (3), and (4), project emissions are summed-up as follows:

$$\begin{aligned}
 PE_y &= PE_{COM,y} + PE_{EC,y} + PE_{FC,y} + PE_{ww,y} \\
 &= 296,261 + 0 + 3,887 + 120,187 = 420,336 \text{ tCO}_2/\text{year}.
 \end{aligned}$$

4.3 Leakage

As per methodology ACM0022, version 03.0, leakage emissions are associated with composting/co-composting, anaerobic digestion and the use of RDF/SB that is exported outside the project boundary. Since the project activity uses incineration technology to treat MSW, leakage emissions are not considered. Therefore, LE_y = 0.

4.4 Net GHG Emission Reductions and Removals

Reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{[Equation 36 of ACM0022]}$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂e/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

LE_y = Leakage emission in year y (t CO₂e/yr)

The results of the emission reduction estimation of project activities in the first crediting period are shown in the table below:

Table 13. Results of emission reduction of the project activity

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
24/07/2022 – 23/07/2023	615,925	420,336	0	195,589
24/07/2023 – 23/07/2024	804,888	420,336	0	384,553
24/07/2024 – 23/07/2025	934,075	420,336	0	513,739
24/07/2025 – 23/07/2026	1,023,022	420,336	0	602,687
24/07/2026 – 23/07/2027	1,084,837	420,336	0	664,502
24/07/2027 – 23/07/2028	1,128,317	420,336	0	707,981
24/07/2028 – 23/07/2029	1,159,367	420,336	0	739,032
Total	6,750,431	2,942,349	0	3,808,082

If the sum of PE_y and LE_y is smaller than 1 per cent of BE_y in the first full operation year of a crediting period, the project participants may choose to assume a fixed percentage of 1 per cent for the sum of PE_y and LE_y for the remaining years of the crediting period.

In the case that overall negative emission reductions arise in a year, CERs are not issued to project participants for the year concerned and in subsequent years, until emission reductions from subsequent years have compensated the quantity of negative emission reductions from the year concerned.

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	GWP _{CH4}
Data unit	t CO ₂ e/t CH ₄
Description	Global Warming Potential for methane
Source of data	Established by Kyoto Protocol
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	IPCC Fifth Assessment report AR5 on Climate Change (2014)
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	GWP _{N2O}
Data unit	t CO ₂ e/t N ₂ O
Description	Global Warming Potential for nitrous oxide
Source of data	Established by Kyoto Protocol
Value applied	265
Justification of choice of data or description of measurement methods and procedures applied	IPCC Fifth Assessment report AR5 on Climate Change (2014)
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	-

Data / Parameter	FFC _j
Data unit	%
Description	Fraction of fossil carbon in total carbon content of waste type j
Source of data	2019 Refinement to the 2006 IPCC Guidelines, volume 5, chapter 2, section 2.3.1 (referred to Table 2.4, chapter 2, volume 5 of IPCC 2006 guidelines)

Value applied	Waste type j	FFC _j
	Paper/cardboard	5
	Textiles	50
	Food waste	-
	Wood	-
	Garden and Park waste	0
	Nappies	10
	Rubber and Leather	20
	Plastic	100
	Metal	NA
	Glass	NA
	Other, inert waste	100
Justification of choice of data or description of measurement methods and procedures applied	Default value	
Purpose of Data	Calculation of project emission	
Comments	This parameter is fixed ex-ante for the entire crediting period.	

Data / Parameter	FCC _j	
Data unit	%	
Description	Fraction of total carbon content of waste type j	
Source of data	2019 Refinement to the 2006 IPCC Guidelines, volume 5, chapter 2, section 2.3.1 (referred to Table 2.4, chapter 2, volume 5 of IPCC 2006 guidelines)	
Value applied	Waste type j	FCC _j
	Paper/cardboard	50
	Textiles	50
	Food waste	50
	Wood	54
	Garden and Park waste	55
	Nappies	90
	Rubber and Leather	67

	Plastic	85
	Metal	NA
	Glass	NA
	Other, inert waste	5
Justification of choice of data or description of measurement methods and procedures applied	Default value	
Purpose of Data	Calculation of project emission	
Comments	This parameter is fixed ex-ante for the entire crediting period.	

Data / Parameter	EF _{CH4,t}				
Data unit	t CH ₄ / t waste (wet basis)				
Description	Emission factor for CH ₄ associated with incineration				
Source of data	2019 Refinement to the 2006 IPCC Guidelines, volume 5, chapter 5, table 5.3				
Value applied	1.21x0.2x10 ⁻⁶				
Justification of choice of data or description of measurement methods and procedures applied	If country – specific data are not available, default values are used as follow:				
	Waste type	Type of incineration/technology		EF _{CH4,t}	
		MSW	Continuous incineration	Stoker	1.21x0.2x10 ⁻⁶
				Fluidised bed	0
			Semi-continuous incineration	Stoker	1.21x6x10 ⁻⁶
				Fluidised bed	1.21x188x10 ⁻⁶
		Batch type incineration	Stoker	1.21x60x10 ⁻⁶	
			Fluidised bed	1.21x237x10 ⁻⁶	
	Industrial sludge (semi-continuous or batch type incineration)			1.21x9700x10 ⁻⁶	
Waste oil (semi-continuous or batch type incineration)			1.21x560x10 ⁻⁶		
The project activity involves continuous stoker-type incineration of MSW, then the value of 1.21x0.2x10 ⁻⁶ is used.					

Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	EF _{N2O,t}		
Data unit	t N ₂ O/ t waste (wet basis)		
Description	Emission factor for N ₂ O associated with incineration		
Source of data	Table 5.6, chapter 5, volume 5 of IPCC 2006 guidelines		
Value applied	1.21x50x10 ⁻⁶		
Justification of choice of data or description of measurement methods and procedures applied	If country – specific data are not available, default values are used as follow:		
	Waste type	Type of management practice	EF _{N2O,t}
	MSW	Continuous and semi-continuous incinerator	1.21x50x10 ⁻³
	MSW	Batch type incineration	1.21x60x10 ⁻³
	Industrial waste	All types of incineration	1.21x100x10 ⁻³
	Sludge (except sewage sludge)	All types of incineration	1.21x450x10 ⁻³
	Sewage sludge	Incineration	1.21x900x10 ⁻³
	The project activity involves continuous stoker-type incineration of MSW, then the value of 1.21x50x10 ⁻³ is applied.		
Purpose of Data	Calculation of project emissions		
Comments	-		

Data / Parameter	B ₀
Data unit	t CH ₄ / t COD
Description	Maximum methane producing capacity, expressing the maximum amount of CH ₄ that can be produced from a given quantity of chemical oxygen demand (t CH ₄ /tCOD)
Source of data	Section 6.2.3.2, chapter 6, volume 5 of IPCC 2006 guidelines
Value applied	0.25

Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	MCF _{ww}
Data unit	-
Description	Methane conversion factor
Source of data	2019 Refinement to the 2006 IPCC Guidelines, volume 5, chapter 6, table 6.3
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	Since country and project specific data are not available, IPCC default values are applied.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	ϕ_y			
Data unit	-			
Description	The model correction factor to account for the model-uncertainties			
Source of data	Tool04 “Emission from solid waste disposal sites”, version 08.0, page 13 of 25			
Value applied	0.85			
Justification of choice of data or description of measurement methods and procedures applied	For baseline emissions: refer to the table below to identify the appropriate factor based on the application of the tool (A or B) and the climate where the SWDS is located. <table><tr><td></td><td>Humid condition/ wet conditions</td><td>Dry conditions</td></tr></table>		Humid condition/ wet conditions	Dry conditions
	Humid condition/ wet conditions	Dry conditions		

	Application A	0.75	0.75
	Application B	0.85	0.8
	The plant is a MSW incineration plant which avoids the disposal of MSW at Nam Son landfill, Hanoi. With annual average temperature (MAT) and rainfall (MAP) are 25.9°C and 1,311.4 mm ³³ respectively, climate conditions in Hanoi falls in tropical climate, MAP > 1000 mm and wet condition. Thus, application B and the climate of wet condition is applied for selecting this value.		
Purpose of Data	Calculation of baseline emissions		
Comments	-		

Data / Parameter	OX
Data unit	-
Description	Oxidation factor, reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste.
Source of data	IPCC 2006 guidelines
Value applied	0.1
Justification of choice of data or description of measurement methods and procedures applied	The value for managed landfill is used. The value is also conservative.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the landfill gas (volume fraction)
Source of data	IPCC 2006 Guidelines

³³ GSO, 2019

Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.5 is recommended by IPCC
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	DOC _f
Data unit	Weight fraction
Description	The fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	IPCC 2006 Guidelines
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	DOC _f value depend on many factors like temperature, moisture, pH, composition of waste, etc. The recommended default value for DOC _f is 0.5. (IPCC 2006 Volume 5, page 3.13)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF
Data unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines
Value applied	1.0
Justification of choice of data or description of measurement methods and procedures applied	Use the following values for MCF: (a) 1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e., waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material;

	(ii) mechanical compacting; or (iii) levelling of the waste. (b) 0.5 for semi-aerobic managed solid waste disposal sites. These must have controlled placement of waste and will include all of the following structures for introducing air to waste layer: (i) permeable cover material; (ii) leachate drainage system; (iii) regulating pondage; and (iv) gas ventilation system. (c) 0.8 for unmanaged solid waste disposal sites – deep and/or with high water table. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of greater than or equal to 5 meters and/or high water table at near ground level. Latter situation corresponds to filling inland water, such as pond, river or wetland, by waste. (d) 0.4 for unmanaged-shallow solid waste disposal sites. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of less than 5 metres. As mentioned in section 1.1, the baseline landfill is an anaerobic managed landfill with a depth of greater 9 meter, and equipped with mechanical compacting, levelling of the waste, and cover material of soil. Therefore, the value of 1.0 is used.
Purpose of Data	Calculation of baseline emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	DOC _j										
Data unit	-										
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)										
Source of data	IPCC 2006 Guidelines (adapted from Volume 5, Chapter 2, Tables 2.4 and 2.5)										
Value applied	The following values are applied: <table border="1"> <thead> <tr> <th>Waste type j</th><th>DOC_j (% wet waste)</th></tr> </thead> <tbody> <tr> <td>Wood and wood products</td><td>43</td></tr> <tr> <td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr> <tr> <td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr> <tr> <td>Textiles</td><td>24</td></tr> </tbody> </table>	Waste type j	DOC _j (% wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverages and tobacco (other than sludge)	15	Textiles	24
Waste type j	DOC _j (% wet waste)										
Wood and wood products	43										
Pulp, paper and cardboard (other than sludge)	40										
Food, food waste, beverages and tobacco (other than sludge)	15										
Textiles	24										

	Garden, yard and park waste	20
	Glass, plastic, metal, other inert waste	0
Justification of choice of data or description of measurement methods and procedures applied	Using 2006 IPCC Guidelines for default value	
Purpose of Data	Calculation of baseline emission	
Comments	This parameter is fixed ex-ante for the entire crediting period.	

Data / Parameter	k _j		
Data unit	1/yr		
Description	Decay rate for the waste type j		
Source of data	IPCC 2006 Guidelines (adapted from Volume 5, Chapter 3, Table 3.3)		
Value applied			
	Waste type j		Tropical (MAT>20°C)- Wet (MAP>1000mm)
	Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.07
		Wood, wood products and straw	0.035
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.17
	Rapidly degrading	Food, food waste, beverages, tobacco (other than sludge)	0.40
	Notes		

	MAT: Mean annual temperature MAP: Mean annual precipitation
Justification of choice of data or description of measurement methods and procedures applied	With annual average temperature (MAT) and rainfall (MAP) are 25.91°C and 1,311.4 mm ³⁴ respectively, climate conditions in Hanoi falls in tropical climate, MAP > 1000 mm and wet condition. The values shown above correspond to these conditions.
Purpose of Data	Calculation of baseline emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor of national grid applying for the plant in year y
Source of data	Data was published by DNA Viet Nam.
Value applied	0.84585
Justification of choice of data or description of measurement methods and procedures applied	The $EF_{grid,CM,y}$ is calculated using $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ which was published by DNA of Viet Nam and weightings w_{OM} , w_{BM} of 0.5 and 0.5 for this project as per “Tool to calculate the emission factor for an electricity system” – Version 07.0
Purpose of Data	Calculation of baseline emission and project emission
Comments	This parameter is fixed ex-ante for the entire crediting period.

5.2 Data and Parameters Monitored

Data / Parameter	$RATE_{compliance,t,y}$
Data unit	Fraction
Description	Rate of compliance with a regulatory requirement to implement the alternative waste treatment t implemented in the project activity

³⁴ GSO, 2019

Source of data	Studies and official reports, such as annual reports provided by municipals bodies
Description of measurement methods and procedures to be applied	Fraction is calculated as the number of instances of compliance divided by the number of instances of compliance plus non-compliance. For ex-ante calculation, the value of 0% is applied. For ex-post calculation, the compliance ratio $RATE_{compliance,y}$ shall be monitored based on the official reports, for instance annual reports provided by municipal bodies. If the rate exceeds 50%, no CERs can be claimed.
Frequency of monitoring/recording	Annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	Fraction is calculated as the number of instances of compliance divided by the number of instances of compliance plus non-compliance.
Comments	Till date, there are no regulations or policies in Vietnam require treating MSW, hence the compliance rate is picked up as 0%.

Data / Parameter	$EFF_{COM,c,y}$
Data unit	Fraction
Description	Combustion efficiency of combustor c in year y
Source of data	2019 Refinement to the 2006 IPCC Guidelines, Volume 5, Chapter 5, page 5.13
Description of measurement methods and procedures to be applied	The source of data shall be the following, in order of preference: 1. Project specific data; 2. Country specific data; or 3. IPCC default values Since the project specific data and country specific data are not available and difficult to obtain, the IPCC default value is applied as a conservative approach.

Frequency of monitoring/recording	Annually
Value applied	100%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	IPCC default value shall be used only when country or project specific data are not available or difficult to obtain.

Data / Parameter	$Q_{\text{waste},c,y} (W_x)$
Data unit	t
Description	Quantity of fresh waste fed into combustor c in year y
Source of data	Project participants
Description of measurement methods and procedures to be applied	Measured with calibrated scales when each waste truck enters the incineration plant gate. The data will be aggregated annually and each monitoring period.
Frequency of monitoring/recording	Continuously, aggregated at least annually
Value applied	1,460,000
Monitoring equipment	Scales
QA/QC procedures to be applied	Scales will be subject to periodic calibration in according to manufacturers' recommendations and comply with Viet Nam regulation.
Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-
Comments	The parameter is equivalent to W_x in calculation of baseline emissions.

Data / Parameter	$P_{n,j,y} / P_{n,j,x}$																								
Data unit	Weight fraction																								
Description	Fraction of waste type j in sample n collected during the year y																								
Source of data	Project participants																								
Description of measurement methods and procedures to be applied	Sample the waste composition, using the waste type j, as provided in table below and weigh each waste fraction (measure on wet basis) Samples of the waste prevented from disposal will be collected and analyzed in the internal laboratory on monthly basis																								
Frequency of monitoring/recording	A minimum of three samples shall be undertaken every three months with the mean value valid for year y																								
Value applied	<table border="1"> <thead> <tr> <th>Waste type j</th><th>$P_{n,j}$</th></tr> </thead> <tbody> <tr> <td>Paper and Cardboard</td><td>2.7 %</td></tr> <tr> <td>Textiles</td><td>1.6%</td></tr> <tr> <td>Food waste</td><td>51.9 %</td></tr> <tr> <td>Wood and wood products</td><td>0%</td></tr> <tr> <td>Garden and park waste</td><td>0%</td></tr> <tr> <td>Nappies</td><td>0%</td></tr> <tr> <td>Rubber and leather</td><td>1.3%</td></tr> <tr> <td>Plastic</td><td>3.0%</td></tr> <tr> <td>Metal</td><td>0.9%</td></tr> <tr> <td>Glass</td><td>0.5%</td></tr> <tr> <td>Others</td><td>38.1%</td></tr> </tbody> </table>	Waste type j	$P_{n,j}$	Paper and Cardboard	2.7 %	Textiles	1.6%	Food waste	51.9 %	Wood and wood products	0%	Garden and park waste	0%	Nappies	0%	Rubber and leather	1.3%	Plastic	3.0%	Metal	0.9%	Glass	0.5%	Others	38.1%
Waste type j	$P_{n,j}$																								
Paper and Cardboard	2.7 %																								
Textiles	1.6%																								
Food waste	51.9 %																								
Wood and wood products	0%																								
Garden and park waste	0%																								
Nappies	0%																								
Rubber and leather	1.3%																								
Plastic	3.0%																								
Metal	0.9%																								
Glass	0.5%																								
Others	38.1%																								
Monitoring equipment	-																								
QA/QC procedures to be applied	The characteristics of different type of waste will be determined a sample randomly with size from 200-250 kg taken during one operating day. Sampling technique ensure the sample is representative. The sample will be mixed, divided into 4 equal parts, and then discard 2 parts and take 2 opposite parts (step 1). These parts will be mixed and repeat step 1. The remaining parts will be mixed and classified into components: (1) Paper/cardboard; (2) Textile; (3) Food waste; (4) Wood; (5) Garden and Park waste; (6) Nappies; (7) Rubber and Leather; (8) Plastics; (9) Metal and glass and (10) Others. The mass as well as the percentage of each portion will be determined and recorded manually.																								
Purpose of data	Calculation of baseline emissions and project emissions																								
Calculation method	-																								

Comments	-
Data / Parameter	Z_y/Z_x
Data unit	-
Description	Number of samples collected during the year y
Source of data	Project participants
Description of measurement methods and procedures to be applied	It is not used in the ex-ante calculation and will be monitored ex-post.
Frequency of monitoring/recording	Minimum of three samples every three months
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	Samples will be taken randomly with size from 200-250 kg taken during one operating day. Sampling technique ensure the sample is representative. The sample will be mixed, divided into 4 equal parts, and then discard 2 parts and take 2 opposite parts (step 1). These parts will be mixed and repeat step 1. The remaining parts will be mixed and classified into components: (1) Paper/cardboard; (2) Textile; (3) Food waste; (4) Wood; (5) Garden and Park waste; (6) Nappies; (7) Rubber and Leather; (8) Plastics; (9) Metal and glass and (10) Others. The mass as well as the percentage of each portion will be determined and recorded manually.
Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-
Comments	-

Data / Parameter	$EC_{PJ,J,y}$
Data unit	MWh
Description	Electricity consumption from the grid as a result of the alternative waste treatment option in year y

Source of data	Direct measurement at the power meters
Description of measurement methods and procedures to be applied	Two-way power meters will be installed at the grid-connected point to measure the amount of electricity imported by the plant from the national electricity grid.
Frequency of monitoring/recording	Continuously, recorded monthly and aggregated annually
Value applied	0 It is assumed as zero and will be updated according to the data monitored
Monitoring equipment	Power meter
QA/QC procedures to be applied	Electricity meter will be calibrated periodically according to national standards and regulations.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$EC_{BL,k,y}$ – SDG7
Data unit	MWh
Description	Electricity generated by the alternative waste treatment option t and exported to the grid in year y
Source of data	Direct measurement at the power meters
Description of measurement methods and procedures to be applied	Two-way power meters will be installed at the grid-connected point to measure the amount of electricity supplied by the plant to the national electricity grid.
Frequency of monitoring/recording	Continuously measurement and at least monthly recording, aggregated annually
Value applied	388,050
Monitoring equipment	Power meters
QA/QC procedures to be applied	Electricity meters will be calibrated periodically according to national standards and regulations. Data on electricity output will be cross-checked with electricity invoices

Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$TDL_{j,y}$ and $TDL_{k,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j or k in year y
Source of data	Tool05, version 03.0
Description of measurement methods and procedures to be applied	Accurate and reliable data within the host country is not available, therefore, the default value from the tool is used. The project activity is meet to scenario A, then the above values will be applied.
Frequency of monitoring/recording	Annually. In the absence of data from the relevant year, most recent figures should be used, but not older than 5 years
Value applied	$TDL_{j,y}$ (for project emission): 20% $TDL_{k,y}$ (for baseline emission): 3%
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-
Comments	-

Data / Parameter	$Q_{ww,y}$
Data unit	m^3
Description	Amount of run-off wastewater generated by the project activity and treated anaerobically or released untreated from the project activity in year y
Source of data	Measured value by flow meter
Description of measurement methods	-

and procedures to be applied	
Frequency of monitoring/recording	Monthly, aggregated annually
Value applied	613,200
Monitoring equipment	Flow meters
QA/QC procedures to be applied	The monitoring instruments will be subject to regular maintenance and testing to ensure accuracy
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	If the wastewater is treated aerobically, emissions are assumed to be zero, and hence this parameter does not need to be monitored

Data / Parameter	$P_{COD,y}$
Data unit	tCOD/m ³
Description	COD of the run-off wastewater generated by the project activity in year y
Source of data	Measured value by purity meter or COD meter
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Monthly, aggregated annually
Value applied	0.035
Monitoring equipment	-
QA/QC procedures to be applied	The monitoring instruments if used will be subject to regular maintenance and testing to ensure accuracy
Purpose of data	Calculation of project emissions
Calculation method	-

Comments	If the run-off wastewater is treated aerobically, emissions are assumed to be zero, and hence this parameter does not need to be monitored
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Data / Parameter	f_y
Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	Local report or written information from operator of the solid waste disposal site and/ or site visit at the solid waste disposal site
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	-

Data / Parameter	$NCV_{i,y}$
Data unit	GJ per mass or volume unit (i.e. TJ/Gg)
Description	Weighted average net calorific value of fuel type i in year y
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Section 1.4, Upper limit.
Description of measurement methods	For ex-ante calculation, the IPCC default value is applied.

and procedures to be applied	For ex-post calculation, following data will be used in order if they are available: (a) Values provided by the fuel supplier in invoices; (b) Measurements by the project participants; (c) Regional or national default values; (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided For (a) and (b): Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	For (a) and (b): The NCV should be obtained for each fuel delivery, from which weighted average annual values should be calculated. For (c): Review appropriateness of the values annually. For (d): Any future revision of the IPCC Guidelines should be taken into account
Value applied	43.3
Monitoring equipment	-
QA/QC procedures to be applied	Verify if the values under (a), (b) and (c) are within the uncertainty range of the IPCC default values as provided in Table 1.2, Vol. 2 of the 2019 Refinement to the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories in (a), (b) or (c) should have ISO17025 accreditation or justify that they can comply with similar quality standards.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	EF _{CO2,i,y}
Data unit	t CO ₂ / GJ
Description	CO2 emission factor per unit of energy of the fuel i
Source of data	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 1, Section 1.4, Upper limit.

Description of measurement methods and procedures to be applied	For ex-ante calculation, the IPCC default value is applied. For ex-post calculation, following data will be used in order if they are available: (a) Values provided by the fuel supplier in invoices; (b) Measurements by the project participants; (c) Regional or national default values; (d) IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided For (a) and (b): Measurements should be undertaken in line with national or international fuel standards
Frequency of monitoring/recording	Annually or Any future revision of the IPCC Guidelines should be taken into account
Value applied	74,800
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$FC_{DO,t,y}$
Data unit	Mass or volume unit per year (i.e ton/yr)
Description	Quantity of fuel type i combusted in process j during the year y
Source of data	On-site measurement – Log book
Description of measurement methods and procedures to be applied	Use either mass or volume meters. In cases where fuel is supplied from small daily tanks, rulers can be used to determine mass or volume of the fuel consumed, with the following conditions: The ruler gauge must be part of the daily tank and calibrated at least once a year and have a book of control for recording the measurements (on a daily basis or per shift). The values will be obtained from log book data maintained by the plant personnel. For conservative, the DO consumption data are aggregated from all the fuel purchase invoices at the project site.

Frequency of monitoring/recording	Continuously and aggregated annually
Value applied	1,200
Monitoring equipment	Ruler gauge. Quantity of DO consumed by the project activity in the year y will be aggregated from DO purchase invoices.
QA/QC procedures to be applied	The consistency of metered fuel consumption quantities in logbook should be cross-checked by an annual energy balance that is based on purchased quantities and stock changes. The metered fuel consumption quantities should also be cross-checked with available purchase invoices from the financial records.
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	The ruler gauge is available as a part of the daily tank. However, quantity of fuel consumed aggregated from all fuel purchase invoices are used as fuel consumption quantity for conservative,

Data / Parameter	SDG8 – Indicator 8.5
Data unit	Number of employees Number of trainings
Description	Number of jobs created during the project activity, trainings conducted and average salary
Source of data	List of employees, Training records, labour contracts, salary records
Description of measurement methods and procedures to be applied	Not applicable
Frequency of monitoring/recording	Once per monitoring period
Value applied	Number of employees: 160 ³⁵ Internal training: Once per year
Monitoring equipment	Not applicable

³⁵ EIA Report, page 181

QA/QC procedures to be applied	Cross-check via interview
Purpose of data	Contribution of the project to sustainable development
Calculation method	Not applicable
Comments	Not applicable

Data / Parameter	SDG 13: Climate action
Data unit	tCO ₂ e/year
Description	The project activity helps to avoid methane release from SWDS(s) in the absence of the project activity and generate renewable electricity, which displaced part of the electricity otherwise supplied by grid-connected fossil fuel fired power plants.
Source of data	Refer to calculation in section 4 of the VCS PD.
Description of measurement methods and procedures to be applied	As per methodology ACM0022 – Alternative waste treatment processes, version 3.0.
Frequency of monitoring/recording	Once per monitoring period
Value applied	Average 544,012 tCO ₂ e annually
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Cross-check with electricity data and domestic solid waste data
Purpose of data	Emission reduction is generated from project activity.
Calculation method	-
Comments	Not applicable

5.3 Monitoring Plan

The monitoring plan is developed in accordance with the procedures for VCS project activity and for the project being implemented. The monitoring plan, which will be implemented by the project participant describes about the monitoring equipment, monitoring organizational structure, policies for oversight and responsibility of personnel, monitoring procedure, parameters to be monitored,

calibration of monitoring equipment, quality assurance/ quality control procedures, data storage and archiving procedures, emergency procedures, training.

A. Monitoring equipment

The main monitoring equipment in the plant will be as follow:

- Weighbridges including four weighbridges with capacity of 50 tons/time and one weigh bridge with capacity of 80 tons/time will be installed at the entrance of the plant. The weight bridge will record the weight of each waste truck going in the plant. Data will be recorded continuously, aggregated daily and monthly. The project owner takes responsibility for the operation and maintenance of the weight bridge in accordance with manufacturer's technical specifications and requirements.
- Wastewater flow meters will be installed at before wastewater treatment system to measure amount of wastewater treated anaerobically from the project activity.
- The electricity delivered to and imported from the national grid will be measured by power meters.

Power meters are installed at the connection point at bus-bar 110 kV of 220 kV Soc Son transformer station. The installed electricity meters are digital bi-direction meters with allowed errors of " $\pm 0.2s$ " for main meter and " $\pm 0.5s$ " for backup meters. Electricity metering system includes main metering system and two backup metering systems are installed according to "Technical design agreement of power metering system, data collection and measurement system of Soc Son Waste to power plant", No.1927/EPTC-KT&CNTT-KDBD dated 09/05/2019 between EVN and the Project Owner in accordance with the electrical metering regulations.

Before the power metering equipment puts into operation, the PO and EVN shall check and accept it. Each terminal block of these equipment is sealed with lead to prevent all the unallowable interferences.

The power meter type used is the electronic 3 phases:

Item	Main meter	1 st backup meters	2 nd backup meters
Type	Three phases, 4 wires	Three phases, 4 wires	Three phases, 4 wires
Model	Elster – A1700	Elster – A1700	Elster – A1700
Manufacturer	UK	UK	UK
Accuracy	0.2s	0.5s	0.5s

B. Organizational structure, responsibilities of personnel, policies for oversight and accountability of monitoring activities

The Project Owner has established a special VCS group to take charge of data collection, supervision, recording and data storage. VCS group is trained and get technical support by the VCS consultation organization.

The structure of monitoring group and the responsibility of members in the VCS Group are the following:

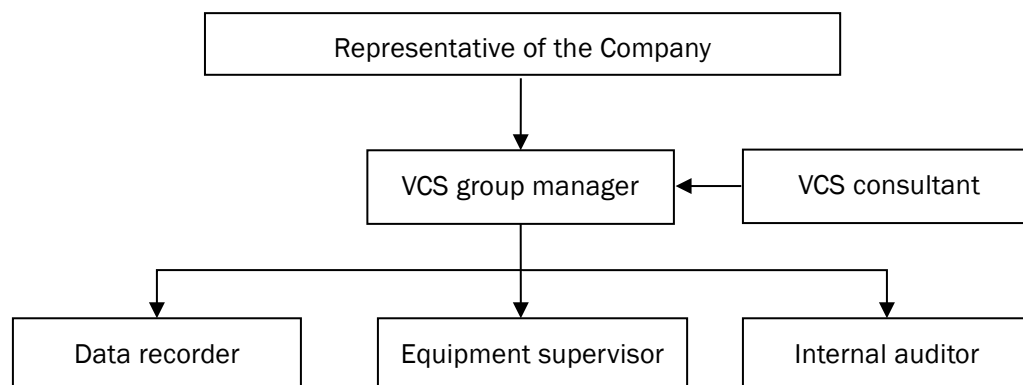


Figure 5: Structure of the monitoring group

The responsibilities of each person involved are elaborated as follows:

Table 14. Group members and their responsibilities

Person	Responsibility
Representative of the Company	Legal representative of Ha Noi Thien Y Environmental Energy JSC. Review the monitoring report annually.
VCS group manager	Managing the whole VCS business of Soc Son waste to power plant project, guiding and supervising data recorder after trained by VCS consultant.
VCS consultant	Providing trainings and technical support about VCS monitoring plan.
Data recorder	Collecting and recording data every month.
Equipment supervisor	Checking equipment periodically according to relevant regulation.
Internal auditor	Checking the monitoring procedures, double checking the collected data.

C. Data collection procedure

Data collection procedure of main parameters are well integrated into MSW management procedure of People's Committee of Ha Noi for amount of input MSW and the standard EVN procedure for electricity generated.

The amount of waste to be incinerated will be measured and recorded automatically when each waste truck enters the incineration plant through the truck scale installed at the gate. The data will be aggregated monthly, and based on the record to calculate the tipping fee. The MSW treatment income receipt will be provided for double check.

The proportion of different type of waste will be determined randomly a sample taken during one month. The sample taken from the waste bunker is classified into (1) Paper/ cardboard; (2) Textile; (3) Food waste; (4) Wood; (5) Garden and Park waste; (6) Nappies; (7) Rubber and Leather; (8) Plastics; (9) Metal and glass; (10) Others. The mass as well as the percentage of each portion will be determined and recorded manually.

The electricity generated and imported from the power grid will be measured continuously by two way power meters installed at the substation. Monthly, EVN staff and staff of the operation

division of the plant will cross-check manual readings from the power meter with the electronically recorded data and sign a joint balance sheet which indicates the amount of power fed into the grid within that month and electricity imported from the grid. This joint balance sheet is also the basis of payment between the EVN and the project proponent. The PO provides electricity sales invoices to EVN as well as receive electricity purchases invoices from EVN.

The type and quantity of auxiliary fuel consumption will be monitored through the provision of fuel purchase invoices.

Other monitored data will be measured with frequency of monitoring as described in section 5.2. Then, data will be collected and recorded monthly by recorders and reported to VCS group manager and provide to VVB during each given verification.

D. Data archiving procedure

- The VCS group appointed by the plant shall keep monitored data in electronic archives at the end of every month. Paper documents should be stored in electronic format.
- All the data and information in the form of paper documents shall be archived by the VCS group, with at least one copy backup for each datum.
- Data monitored and required for verification and issuance are to be kept for two years after the end of the crediting period or the last issuance of VCUs for this project activity, whichever occurs later.

E. The procedures for handling non-conformances with the validated monitoring plan

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible. In doing so, the chief of division shall seek guidance from VCS consultant.

If the weighbridges are broken, the amount of input solid waste will be adopted as minimum historical data for resulting minimum baseline emissions. However, this case will be rarely happened due to 5 weighbridges working in shifts, the case of one broken, the others will work full capacity continuously.

If the wastewater flow meter is broken, the amount of wastewater will be historical data for a maximum project emission.

The main and backup power meters are used in order to record the electricity generated to the grid.

- In normal condition, the data of main power meter will be used as the basis of payment by the EVN to the PO and to calculate the emission reductions by the project activity.
- In case the main power meter is in failure and the backup power meter is still in good operation, the result of the backup meter will be used to calculate the emission reduction by the project activity.
- In case of both main and backup power meters are in failure, the Project Owner and the power company (EVN) will jointly calculate a conservative estimate of power supplied to the grid. The assumptions used to estimate net electricity supply to the grid will be signed by both a representative of the project owner as well as a representative of the power company (EVN).
- In case any power meters are in failure, the Project Owner will inform EVN immediately and contract with the authorized party (belong to EVN) to verify/ calibrate and/or replace the failed power meter.

G. Personnel Training

- In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff (VCS team) will be trained. The plant helpers will be trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.
- During the operation period, the Project Owner will hold irregularly scheduled training to improve staff's professional level
- The new staff is not allowed to operate or maintain the equipment, monitoring system until they master the knowledge and skills after training mentioned above.

H. The procedures for internal auditing and Quality assurance and quality control (QA/QC):

- Internal auditing: This is an internal procedure to ensure the quality of operation and VCS monitoring. The meetings for internal audit are organized every year by the internal auditor. All the internal audits are recorded and stored.
- QA/QC: The monitoring processes and all data collected as part of monitoring plan are double checked by internal auditor and VCS group manager. All the collected data are archived in both electronic and hard copy, and shall be kept until 2 years after the end of the crediting period, or the last issuance of VCUs, whichever is later.

I. The procedures for calibrating monitoring equipment

- Weigh bridges: The weigh bridges will be calibrated and verified comply with Viet Nam current regulation. As per Circular No. 07/2019/TT-BKHCN amendments to circular no. 23/2013/TT-BKHCN dated 26/09/2013 of Ministry of Science and Technology on measurement of group 2 measuring instrument, the calibration of weigh bridges need to be implemented each 12 months.
- Power meters: The meters will be calibrated and verified pursuant to national standard. According to the Decision No. 2739/QĐ-TĐC on “promulgating metrological technical standard of Viet Nam” issued by Directorate for Standards, Metrology and Quality under Ministry of Science and Technology of Viet Nam on 23/12/2019, Article 2, the metrology standard – DLVN 39:2019 takes effect from 01/01/2020. Paragraph 8.3, DLVN 39:2019 – “Alternating current static watt-hour meters – Verification procedure”, the calibration for 3 phase meters need to be conducted every three years by the third party once during project operation. After every calibration, the meters will be sealed so that no illegal interference is possible.
- Wastewater flow meter: Flow meter will be calibrated and verified pursuant to National Standard. As per Circular No. 07/2019/TT-BKHCN amendments to circular no. 23/2013/TT-BKHCN dated 26/09/2013 of Ministry of Science and Technology on measurement of group 2 measuring instrument, the calibration of flow meter need to be implemented each 60 months.
- Fuel meter ruler: The ruler gauge will be calibrated at least once a year as requirement of tool.
- COD meter: The monitoring instruments will be subject to regular maintenance and testing to ensure accuracy as per the manufacturer's instruction.

K. Sampling approaching

Not applicable.

APPENDIX 1

BASELINE INFORMATION

Data sources using to calculate $EF_{grid,CM,y}$ has been referred to the published data of DNA Vietnam according to the link below:

<http://dcc.gov.vn/van-ban-phap-luat/1066/He-so-phat-thai-luoi-dien-Viet-Nam-2019.html>

Data of power plants in the Vietnam national grid in 2017, 2018 and 2019

Table 15: Fuel consumed, emissions and output in the most recent 3 years (2017- 2019)

Plants	Fuel consumed (Coal, Oil: kton; Gas: mm ³)	Grid connected output (MWh)	Emissions (tCO ₂ e)
2017	40,219.86	96,840,719.67	83,160,505.77
Coal-fired	32,592.36	56,558,815.75	65,743,098.48
Gas-Turbin	7,619.00	39,854,916.07	17,387,786.97
Oil-fired	5.91	21,683.55	20,879.21
Diesel DO	2.59	11,157.40	8,741.11
2016	51,886.31	114,997,401.74	106,748,597.11
Coal-fired	44,246.77	75,082,041.30	89,435,646.18
Gas-Turbin	7,627.76	39,772,700.73	17,272,563.05
Oil-fired	9.15	131,212.81	31,646.77
Diesel DO	2.63	11,446.90	8,741.11
2017	61,378.19	143,651,728.86	126,361,614.48
Coal-fired	52,962.33	100,226,598.84	106,428,523.23
Gas-Turbin	8,173.16	42,387,843.20	19,129,513.21
Oil-fired	229.30	984,286.82	760,908.80
Diesel DO	13.4	53,000.00	42,669.24

Table 16: Total emission and power output of the most recent 3 years

	2017	2018	2019	Total
Total output (MWh)	96,446,572.77	114,997,401.74	143,651,728.86	355,095,703.37
Total Emission (tCO ₂ e)	83,160,505.77	106,748,579.11	126,361,614.48	316,270,717.36

Table 17: Calculate the BM emission factor in 2019

Plant	Commission year	Fuel consumed (Coal; Oil: kton; Gas: mm ³)	Grid connected output (MWh)	Emissions (tCO ₂ e)
5 most recent – built plants				
Mui Dinh	2019	Wind power	62,000.00	-
Da Bac 4	2019	Solar power	40,000.00	-
Halcom Solar	2019	Solar power	74,000.00	-
Vinh Tan 4 S3	2019	Coal	897,094.85	916,262.52
Vinh Hao 6	2019	Solar power	55,000.00	-
Total			1,128,094.85	916,262.52
Most recent-built plants charged 20% total output				
Mui Dinh	2019	Wind power	62,000.00	-
Da Bac 4	2019	Solar power	40,000.00	-
Halcom Solar	2019	Solar power	74,000.00	-
Vinh Tan 4 S3	2019	Coal	897,094.85	916,262.52
Vinh Hao 6	2019	Solar power	55,000.00	-
Vinh Tan 2	2019	Solar power	149,000.00	-
Da Bac 3	2019	Solar power	43,000.00	-
Dakre H2	2019	Hydropower	50,500.00	-
Dakre H1	2019	Hydropower	50,500.00	-
Cam Hoa	2019	Solar power	58,000.00	-
AMI Khanh Hoa	2019	Solar power	22,000.00	-
Tuy Phong	2019	Solar power	18,433.74	-
Ho Bau Ngu	2019	Solar power	28,418.70	-
Thuan Minh 2	2019	Solar power	35,870.93	-
Dau Tieng 3	2019	Solar power	63,216.53	-

Gala	2019	Solar power	44,000.00	-
Thinh Long AAA Phu Yen	2019	Solar power	56,000.00	-
Ninh Phuoc 6,1 & 6,2	2019	Solar power	32,092.84	-
Europplast Phu Yen	2019	Solar power	54,000.00	-
Phuoc Huu Dien Luc 1	2019	Solar power	35,184.10	-
Trung Nam Tra Vinh	2019	Solar power	38,000.00	-
Thuan Nam	2019	Solar power	29,381.00	-
Nhi Ha	2019	Solar power	38,000.00	-
Fujiwara	2019	Solar power	55,000.00	-
Solarpark 2	2019	Solar power	25,000.00	-
Binh An	2019	Solar power	25,000.00	-
Solarpark 1	2019	Solar power	187,000.00	-
Son My 3.1	2019	Solar power	42,157.71	-
Van Giao 1	2019	Solar power	125,000.00	-
Van Giao 2	2019	Solar power	51,000.00	-
Phan Lam	2019	Solar power	29,000.00	-
CMX	2019	Solar power	149,411.44	-
Sao Mai	2019	Solar power	86,505.23	-
BCG Bang Duong	2019	Solar power	52,000.00	-
Bach Khoa A Chau	2019	Solar power	54,000.00	-
My Son Hoan Loc Viet	2019	Solar power	38,000.00	-
Dien luc Mien Trung	2019	Solar power	10,063.33	-
Dau Tieng 2	2019	Solar power	206,126.38	-
Xuan Tho 2	2019	Solar power	45,163.22	-
Xuan Tho 1	2019	Solar power	43,166.57	-
Truc Son	2019	Solar power	22,000.00	-

Dau Tieng 1	2019	Solar power	164,585.17	-
Hong Phong 4	2019	Solar power	30,000.00	-
Mui Ne	2019	Solar power	202,000.00	-
VSP Binh Thuan II	2019	Solar power	43,840.00	-
KCN Chau Duc	2019	Solar power	45,000.00	-
Vinh Hao 4	2019	Solar power	32,000.00	-
Tri Viet 1	2019	Solar power	45,163.22	-
Hoa Hoi	2019	Solar power	43,000.00	-
Long Thanh 1	2019	Solar power	87,000.00	-
Hoang Thai Gia HTG	2019	Solar power	174,000.00	-
HCG Tay Ninh	2019	Solar power	43,785.16	-
Europlast Long An	2019	Solar power	44,483.80	-
Cat Hiep	2019	Solar power	57,000.00	-
Phuoc Huu	2019	Solar power	37,000.00	-
Vinh Hao	2019	Solar power	24,219.60	-
LIG Quang Tri	2019	Solar power	45,566.32	-
Ham Kiem	2019	Solar power	10,000.00	-
Da Bac	2019	Solar power	7,680.00	-
Da Mi	2019	Solar power	45,000.00	-
Gelex Ninh Thuan	2019	Solar power	36,677.69	-
BIM 2	2019	Solar power	302,066.00	-
BIM 3	2019	Solar power	57,252.35	-
Trung Nam	2019	Solar power	247,812.20	-
Phong Phu	2019	Solar power	41,679.08	-
Buon Ma Thuot	2019	Solar power	30,328.53	-
BIM	2019	Solar power	28,255.80	-
Cu Jut	2019	Solar power	57,245.72	-

Duc Hue 1	2019	Solar power	51,056.80	-
Trung Nam	2019	Wind Power	247,812.20	-
Song Giang	2019	Solar power	20,000.00	-
TTC no.2	2019	Solar power	36,000.00	-
Ham Phu 2	2019	Solar power	38,000.00	-
Song Luy 1	2019	Solar power	37,000.00	-
Yen Dinh	2019	Solar power	45,000.00	-
TTC no. 1	2019	Solar power	44,000.00	-
BP Solar 1	2019	Solar power	46,000.00	-
Hong Phong 1B	2019	Solar power	33,000.00	-
Quang Minh	2019	Solar power	72,699.32	-
Srepok 1	2019	Solar power	74,085.40	-
Hong Phong 1A	2019	Solar power	38,000.00	-
Da Bac 2	2019	Solar power	13,000.00	-
Da Nhim H5	2018	Hydropower	142,410.10	-
Thang Long S2	2018	Coal	1,858,148.00	1,979,290.03
Thang Long S1	2018	Coal	1,860,331.00	1,977,255.50
Vinh Tan 4 S1	2018	Coal	3,302,372.13	3,536,545.08
H1 Nam Cun	2018	Hydropower	80,682.46	-
H2 Nam Cun	2018	Hydropower	78,216.33	-
An Khe biomass	2018	Bagasse	97,270.00	-
Nho Que H1	2017	Hydropower	54,350.13	-
H1 Bao Lam 3	2017	Hydropower	88,263.02	-
H2 Bao Lam 3	2017	Hydropower	88,263.02	-
S3 Formosa Ha Tinh	2017	Other bituminous coal	569,462.00	780,357.36
KCP Phu Yen	2017	Bagasse	73,719.00	-

S1 Thai Binh	2017	Coal	1,565,133.90	1,636,109.71
Thuan Hoa H1	2017	Hydropower	71,734.00	-
Thuan Hoa H2	2017	Hydropower	82,783.00	-
Vinh Tan 4 S2	2017	Coal	3,163,760.42	3,434,290.79
Thac Mo H3	2017	Hydropower	49,175.00	-
S2 Thai Binh 1	2017	Coal	1,724,856.00	1,806,167.00
S1 Duyen Hai 3	2017	Coal	3,677,516.00	3,462,456.00
S2 Duyen Hai 3	2017	Coal	4,271,974.00	4,045,937.00
S1 Duyen Hai 1	2016	Coal	3,324,689.00	3,329,500.00
S2 Duyen Hai 1	2016	Coal	3,647,511.00	3,655,185.00
S2 Formosa Ha Tinh	2016	Other bituminous coal	569,462.00	780,357.00
Formosa S3	2016	Other bituminous coal	2,093,452.00	1,205,684.00
Huoi Quang H2	2016	Hydropower	946,186.00	-
Lai Chau H2	2016	Hydropower	954,792.00	-
Lai Chau H3	2016	Hydropower	865,132.00	-
Nho Que H2	2016	Hydropower	52,065.00	-
Nong Son	2015	Coal	167,680.76	219,369.45
Viet Nam sugar plant	2015	Bagasse	7.66	-
An Khanh	2015	Coal	769,923.00	1,121,502.86
Total			42,306,901.18	33,886,270.57